Locally Differentially Private Degree List Collection in Streaming Directed Graphs

Ziyao Wei, Qing Liu, Zhikun Zhang *Member, IEEE*, Yunjun Gao, *Senior Member, IEEE*

***Abstract*—The degree list of a graph supports applications such as recommendation, risk assessment, and graph analysis, but collecting users' degrees directly can reveal sensitive information. This problem is particularly challenging in streaming directed graphs, where the degree list changes over time and existing privacy-preserving methods largely focus on static graphs or undirected streams. We study online degree list collection under w-event local differential privacy. We first establish a composition theorem for event-level local differential privacy and use it to design a collection algorithm that satisfies w-event local differential privacy for each user. To ensure that the noisy estimates correspond to a realizable directed graph, we develop a post-processing algorithm that adjusts the estimated degree list to be digraphic. We further propose an optimized algorithm that exploits correlations between snapshots to reduce computation while preserving the privacy guarantee. Experiments demonstrate the effectiveness of the proposed algorithms in terms of utility and efficiency.**

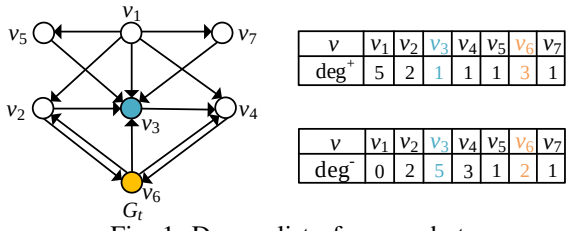
***Index Terms*—degree list, local differential privacy, streaming directed graphs**

I. Introduction

DIRECTED graphs can model asymmetric relationships and have been used in many fields, such as web discovery [1], [2], task scheduling [3], and social network analysis [4]–[8]. In real-world applications, the edges of a directed graph may change constantly, which can be modeled by a streaming directed graph [9], [10] consisting of a sequence of snapshots.

The degree list of a snapshot in a streaming directed graph contains the out-degree and in-degree of each vertex. For example, in Fig. 1, the snapshot G_t's degree list contains two sub-lists. One is the out-degree list, and the other is the in-degree list. The out-degree list (resp. in-degree list) contains all the vertices' out-degrees (resp. in-degrees). Specifically, v_3's out-degree and in-degree are 1 and 5, respectively. v_6's out-degree and in-degree are 3 and 2, respectively.

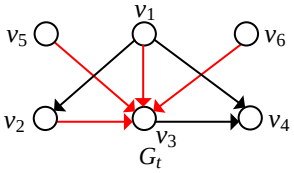
Collecting the degree list of a streaming directed graph can support many applications. For example, social platform usually provides popular users and active users, which can recommend potentially interesting users to each user. One method to obtain these users is to recognize the high out-degree vertices and the high in-degree vertices in user relationship network. For example, in snapshot G_t shown in Fig. 1, user v_3 has the highest in-degree, then we can infer that v_3 is the most popular user. Moreover, user v_1 has the highest out-degree, then we can also infer that v_1 is the most active user. Besides, the degree list collection can support many downstream tasks. For example, graph collection [11], [12]. Therefore, it is necessary to collect the degree lists of streaming directed graphs. However, directly collecting the degree list may lead to privacy leakage, since the central server is usually untrusted. For example, Fig. 2 shows a snapshot G_t. The untrusted central server and all the users have a consensus that the snapshot at each timestamp does not contain self-loops or multiple edges. Notice that in G_t, v_3's in-degree is 4 and v_4's out-degree is 0. After the untrusted central server has collected all the degrees, it can infer that edges (v_1, v_3), (v_2, v_3), (v_5, v_3), and (v_6, v_3) exist. Therefore, the untrusted central server obtains the neighborhood information of v_1, v_2, v_3, v_5, and v_6, resulting in privacy leakage.



v	v_1	v_2	v_3	v_4	v_5	v_6	v_7
$\deg^+$	5	2	1	1	1	3	1

v	v_1	v_2	v_3	v_4	v_5	v_6	v_7
$\deg^-$	0	2	5	3	1	2	1

Fig. 1: Degree list of a snapshot



v	v_1	v_2	v_3	v_4	v_5	v_6
$\deg^+$	3	1	1	0	1	1

v	v_1	v_2	v_3	v_4	v_5	v_6
$\deg^-$	0	1	4	2	0	0

Fig. 2: An example of privacy leakage

Existing works [13]–[17] under privacy-preserving constraints mainly employ differential privacy [18] and focus on static graphs or streaming undirected graphs. However, existing algorithms have two limitations. First, they require the number of timestamps as input, which makes them unsuitable for infinite graph streams. Second, they support only one edge insertion or deletion per timestamp and cannot handle multiple updates at one timestamp. We consider streaming directed graphs in which multiple edges may be inserted or deleted at each timestamp. This batch-update model is supported by widely used open-source systems such as Apache Flink [19] and Apache Spark [20], making privacy in this setting practically important. However, applying existing algorithms to this model can violate differential privacy. We therefore study the online w-event locally differentially private degree list collection problem. Unlike event-level differential

Z. Wei, Q. Liu, Z. Zhang, and Y. Gao are with the College of Computer Science and Technology, Zhejiang University, Hangzhou 310027, China (e-mail: wei_zy@zju.edu.cn; qingliucs@zju.edu.cn; zhikun@zju.edu.cn; gaoyj@zju.edu.cn)

*Corresponding author: Yunjun Gao (e-mail: gaoyj@zju.edu.cn)

privacy [21]–[23], w-event differential privacy [24]–[27] does not limit protection to a single timestamp; unlike user-level differential privacy [28]–[30], it does not suffer from privacy budget exhaustion. Its algorithm design presents several key technical challenges:

Challenge I: How to implement w-event local differential privacy online? The key challenge is to provide an online release at every timestamp while ensuring that the privacy budget used in any window of length w does not exceed the specified limit. A direct application of event-level mechanisms does not by itself ensure this window-based guarantee. Our innovation is to establish a composition theorem that connects event-level local differential privacy with w-event differential privacy. We then use this theorem to construct an online degree list collection algorithm with a formal w-event local differential privacy guarantee for each user.

Challenge II: How to make the released degree list digraphic? The privacy mechanism usually introduces noise, so the released in-degree and out-degree lists may not correspond to any directed graph. Berger's characterization gives necessary and sufficient conditions for a non-negative list to be digraphic [31], but satisfying the conditions jointly is nontrivial: correcting one can violate the other. Our contribution is a post-processing algorithm that exploits the relationship between these conditions to produce a valid digraphic degree list, while changing the noisy estimates as little as possible.

Challenge III: How to determine the releasing method at each timestamp without privacy leakage? Snapshots in a stream are correlated, so reusing a previous estimate can reduce computation. Yet deciding whether to refresh a degree is itself data-dependent; revealing that decision without protection could leak information and invalidate the overall privacy guarantee. Our innovation is a privacy-preserving update-selection technique that determines, for each user's in-degree and out-degree, whether a new estimate is needed while maintaining w-event local differential privacy. This lets the algorithm exploit temporal correlation without violating differential privacy.

To implement locally differentially private degree list collection in streaming directed graphs, we make the following contributions in this paper:

- We develop an online algorithm for collecting degree lists in streaming directed graphs under w-event local differential privacy. We establish a composition theorem for event-level local differential privacy and use it to prove that the algorithm satisfies the desired w-event local differential privacy guarantee for each user.
- We design an efficient post-processing algorithm that adjusts noisy in-degree and out-degree lists to form a valid digraphic degree list. Guided by Berger's necessary and sufficient characterization [31], the algorithm coordinates the two lists while keeping the adjusted estimates close to the noisy ones.
- We identify a sparsity issue in the exponential mechanism used by our w-event locally differentially private algorithms. To mitigate it, we introduce a step size that reduces the mechanism's output range and alleviates the sparsity phenomenon.
- We propose a privacy-preserving update-selection technique that determines whether each user's estimated in-degree and out-degree should be refreshed at each timestamp. The technique exploits temporal correlations to improve efficiency while preserving w-event local differential privacy.
- We evaluate the proposed algorithms in terms of utility and efficiency. The experimental results demonstrate that our methods achieve strong utility while remaining computationally efficient.

II. Related Works

We review the related works from two aspects: differential private degree list release and differential privacy with streaming data.

Differentially Private Degree List Release. For degree list release, there are many differentially private algorithm. Hay et al. [15] proposed a degree list publication algorithm under k-edge differential privacy. Following [15], Karwa and Slavković [16] proposed an algorithm that includes an additional step to ensure that the released degree list is graphical. Under node differential privacy, Raskhodnikova and Smith [17] designed a degree list publication algorithm by computing a low-stretch Lipschitz extension via convex flow optimization, adding calibrated Laplace noise, and adaptively selecting the optimal degree bound using a generalized exponential mechanism. Recently, Raskhodnikova and Steiner [14] proposed algorithms for publishing degree list in streaming undirected graph. However, these algorithms are limited to finite graph streams, with only one edge being added or deleted between consecutive snapshots.

Differential Privacy with Streaming Data. In streaming differential privacy, there are three common used types of differential privacy: (1) event-level differential privacy; (2) user-level differential privacy; (3) w-event differential privacy. For event-level differential privacy, Dwork et al. [32] first studied the event-level differential privacy with streaming data, and Chan et al. [33] first considered the infinite streaming data. Following these works, Henzinger et al. [23], [34], [35] proposed algorithms for counting, sum, and observation. Xie et al. [21] proposed an efficient and accurate cardinality estimation method. Wang et al. [22] designed observation algorithms under both centralized and local differential privacy. For user-level differential privacy, Dong et al. [29] proposed an observation algorithm and then extended it to a join algorithm. Bao et al. [28] devised a collection algorithm with local differential privacy. For w-event differential privacy, Kellaris et al. [24] first defined w-event differential privacy, and designed two observation algorithms. Li et al. [36] proposed two algorithms for observation. For w-event local differential privacy, Ren et al. [26] proposed observation algorithms for infinite streaming data. Du et al. [25] proposed two algorithms for observation, which are different in privacy budget allocation. Liu and Zhao [37] also proposed an observation algorithm.

In contrast to these studies, we address online degree list collection over streaming directed graphs, whether finite or

infinite, where multiple edges may be inserted or deleted at each timestamp.

III. Preliminaries

For any $a, b \in \mathbb{Z} \cup \{\pm\infty\}$ such that $a \leq b$, let $[a, b] = \{x \in \mathbb{Z} : a \leq x \leq b\}$. If $a = 1$, we use $[b]$ to denote $[1, b]$. A streaming directed graph is denoted by a finite or infinite sequence of snapshots, i.e., $\mathbf{G} = (G_t)_{t=1}^{T}$. If $T \in \mathbb{N}$, $\mathbf{G}$ is finite. If $T = +\infty$, $\mathbf{G}$ is infinite. $\forall t \in [T]$, the snapshot at timestamp t is a directed graph, denoted by $G_t = (V_t, E_t)$. Note that all snapshots share the same vertex set, and thus we just use V instead of V_t in the remainder of the paper. Let $n = |V|$ and $m_t = |E_t|$. For a vertex $v_i \in V$, $\mathbf{u}_{i,t}^{+} = (u_{i,t,1}^{+}, u_{i,t,2}^{+}, \cdots, u_{i,t,n}^{+}) \in \{0,1\}^n$ and $\mathbf{u}_{i,t}^{-} = (u_{i,t,1}^{-}, u_{i,t,2}^{-}, \cdots, u_{i,t,n}^{-}) \in \{0,1\}^n$ denote the out-neighbor list and in-neighbor list of v_i at timestamp t, respectively. In particular, if the vertex v_j is an out-neighbor (resp. in-neighbor) of v_i at timestamp t, $u_{i,t,j}^{+} = 1$ (resp. $u_{i,t,j}^{-} = 1$); otherwise, $u_{i,t,j}^{+} = 0$ (resp. $u_{i,t,j}^{-} = 0$). v_i's out-neighbor list and in-neighbor list together constitute its neighbor list, denoted by $\mathbf{u}_{i,t} = (\mathbf{u}_{i,t}^{+}, \mathbf{u}_{i,t}^{-}) \in \{0,1\}^n \times \{0,1\}^n$. If all timestamps are considered, v_i's neighbor lists are denoted by $\mathbf{u}_i \in (\{0,1\}^n \times \{0,1\}^n)^T$. We use $d_{t,i}^{+}$ and $d_{t,i}^{-}$ to denote the in-degree and out-degree of v_i at timestamp t, respectively. Correspondingly, $d_t^{+} = (d_{t,1}^{+}, d_{t,2}^{+}, \cdots, d_{t,n}^{+}) \in [0, n-1]^n$ and $d_t^{-} = (d_{t,1}^{-}, d_{t,2}^{-}, \cdots, d_{t,n}^{-}) \in [0, n-1]^n$ denote the out-degree and in-degree lists of G_t, respectively.[1]

A. w-event Local Differential Privacy

We first define when two neighbor lists are considered w-neighboring, and then use this relation to define w-event local differential privacy.

Definition III.1 (w-neighboring Neighbor Lists)**.** *For $n \in \mathbb{N}$ and $T \in \mathbb{N} \cup \{+\infty\}$, let $w \in \mathbb{N}$ be such that $w \leq T$. Two neighbor lists $\mathbf{u}$, $\mathbf{u}' \in (\{0,1\}^n \times \{0,1\}^n)^T$ are w-neighboring if there exists $\mathcal{I} \in \{[s, s+w-1] : s \in [T-w+1]\}$ such that $\{t \in [T] : \mathbf{u}_t \neq \mathbf{u}'_t\} \subseteq \mathcal{I}$.*

In other words, w-neighboring neighbor lists are two neighbor lists that differ from each other only within a time window of length at most w, and they are identical outside of this window. We call $\mathcal{I}$ a window. Based on it, we introduce w-event local differential privacy.

Definition III.2 (w-event Local Differential Privacy)**.** *For $n \in \mathbb{N}$, $T \in \mathbb{N} \cup \{+\infty\}$, and $\varepsilon \in \mathbb{R}_{\geq 0}$, let $w \in [T]$. A randomized algorithm $\mathcal{A}$ satisfies w-event local differential privacy with privacy budget ε if for any pair of w-neighboring neighbor lists $\mathbf{u}, \mathbf{u}' \in (\{0,1\}^n \times \{0,1\}^n)^T$, and any measurable set $\mathcal{O} \subseteq \mathrm{Range}(\mathcal{A})$,*

$$\Pr[\mathcal{A}(\mathbf{u}) \in \mathcal{O}] \leq e^{\varepsilon} \Pr[\mathcal{A}(\mathbf{u}') \in \mathcal{O}].$$

We next recall the exponential mechanism, a standard mechanism for achieving differential privacy. Sections IV and V show how we use it to implement w-event local differential privacy.

[1]When $T = +\infty$, $(\{0,1\}^n \times \{0,1\}^n)^T := (\{0,1\}^n \times \{0,1\}^n)^{\mathbb{N}}$.

Definition III.3 (Global Sensitivity [38])**.** *Let $f : \mathcal{U} \times \mathcal{R} \to \mathbb{R}$ be a utility function. The global sensitivity of f is defined by,*

$$GS_f := \sup_{r \in \mathcal{R}} \sup_{\mathbf{u} \sim \mathbf{u}' : \mathbf{u}, \mathbf{u}' \in \mathcal{U}} |f(\mathbf{u}, r) - f(\mathbf{u}', r)|,$$

where $\mathbf{u} \sim \mathbf{u}'$ means that $\mathbf{u}$ and $\mathbf{u}'$ are neighboring.

Theorem III.4 (Exponential Mechanism [38])**.** *Let $\mathcal{A} : \mathcal{U} \to \mathcal{R}$ be a randomized algorithm, $f : \mathcal{U} \times \mathcal{R} \to \mathbb{R}$ be a utility function, $\mathbf{u} \in \mathcal{U}$ be an input, and $\varepsilon \in \mathbb{R}_{\geq 0}$ be a privacy budget. If for any $r \in \mathcal{R}$, $\mathcal{A}(\mathbf{u}) = r$ holds with probability proportional to $\exp(\frac{\varepsilon f(\mathbf{u}, r)}{2GS_f})$, then $\mathcal{A}$ satisfies differential privacy with privacy budget ε.*

Proposition III.5 ([38])**.** *Let $f : \mathcal{U} \times \mathcal{R} \to \mathbb{R}$ be a utility function, where $\mathcal{R}$ is a finite set. For any $\varepsilon \in \mathbb{R}_{\geq 0}$, let $\mathcal{A} : \mathcal{U} \to \mathcal{R}$ be an exponential mechanism with privacy budget ε. For any $\beta \in \mathbb{R}_{\geq 0}$, $\mathbf{u} \in \mathcal{U}$, we have,*

$$\Pr[f(\mathbf{u}, \mathcal{A}(\mathbf{u})) \leq \max_{x \in \mathcal{R}} f(\mathbf{u}, x) - \frac{2GS_f}{\varepsilon}(\log|\mathcal{R}| + \beta)] \leq \exp(-\beta).$$

Differential privacy has some nice properties, including compositions and post-processing.

Theorem III.6 (Basic Composition [39])**.** *Let $\mathcal{A}_1 : \mathcal{D} \to \mathcal{R}_1$ be an algorithm satisfying differential privacy with privacy budget ε_1, and let $\mathcal{A}_2 : \mathcal{D} \to \mathcal{R}_2$ be an algorithm satisfying differential privacy with privacy budget ε_2. Then, the basic composition $(\mathcal{A}_1, \mathcal{A}_2) : \mathcal{D} \to \mathcal{R}_1 \times \mathcal{R}_2$ satisfies differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2)$.*

Theorem III.7 (Adaptive Composition [39])**.** *Let $\mathcal{A}_1 : \mathcal{D} \to \mathcal{R}_1$ be an algorithm satisfying differential privacy with privacy budget ε_1, and let $\mathcal{A}_2 : \mathcal{R}_1 \times \mathcal{D} \to \mathcal{R}_2$ be an algorithm satisfying differential privacy with privacy budget ε_2 for any fixed $r_1 \in \mathcal{R}_1$. Then, the adaptive composition $(\mathcal{A}_1, \mathcal{A}_2) : \mathcal{D} \to \mathcal{R}_1 \times \mathcal{R}_2$ satisfies differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2)$.*

Theorem III.8 (Parallel Composition [40], [41])**.** *Let $\mathcal{A}_1$, $\mathcal{A}_2$, $\cdots$, $\mathcal{A}_k$ be algorithms depending on disjoint subsets of input datasets. If for any $i \in [k]$, $\mathcal{A}_i$ satisfies differential privacy with privacy budget ε_i, then the parallel composition $(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_k)$ satisfies differential privacy with privacy budget $\max_{i \in [k]} \varepsilon_i$.*

Theorem III.9 (Post-Processing [38])**.** *Let $\mathcal{A}$ be an algorithm satisfying differential privacy with privacy budget ε. For any measurable mapping f, $f \circ \mathcal{A}$ satisfies differential privacy with privacy budget ε.*

B. System Model and Threat Model

As shown in Fig. 3, given a streaming directed graph $\mathbf{G}$, the system model contains an untrusted central server, and each vertex v_i in $\mathbf{G}$ corresponds to a user. Each user v_i holds a view of v_i's out-neighborhoods and in-neighborhoods at each timestamp. If user v_1 is an out-neighborhood of user v_2, then user v_2 is an in-neighborhood of user v_1. In Fig. 3, at timestamp t_1, edges (v_3, v_1), (v_3, v_2), and (v_5, v_2) exist. At

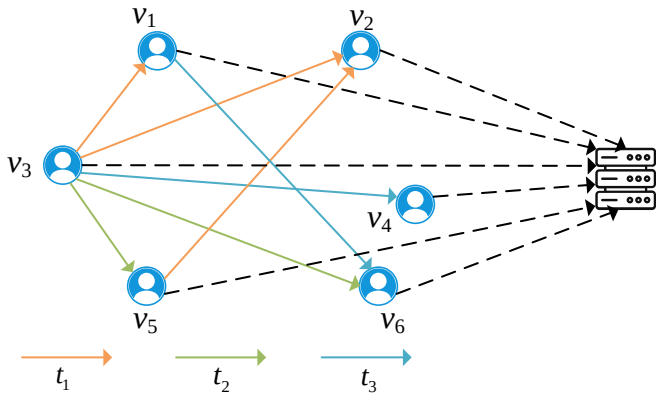


Fig. 3: System model

timestamp t_2, edges (v_3, v_5) and (v_3, v_6) exist. At timestamp t_3, edges (v_1, v_6) and (v_3, v_4) exist.

Threat Model. The central server is honest-but-curious: it follows the protocol but may attempt to infer sensitive information about users from the collected data. Specifically, the central server aims to infer the out-neighborhoods and in-neighborhoods of each user v_i. To protect privacy, each user does not release the exact information about neighborhoods to the central server. Instead, each user applies w-event local differential privacy algorithm when releasing the information about neighborhoods to the central server. The goal is to prevent the untrusted central server from accurately inferring any user's out-neighborhoods and in-neighborhoods while collecting some information from each user.

C. Problem Definition

To support the real-world application, we aim to release the degree list online, thus at each timestamp $t \in [T]$, the algorithm should release the degree list $d_t = (d_t^+, d_t^-) \in [0, n-1]^n \times [0, n-1]^n$ of the snapshot G_t.

Problem III.10. *Let $\mathbf{G}$ be a streaming directed graph containing T timestamps. The goal is to find an algorithm $\mathcal{A}$, such that at each timestamp $t \in [T]$, $\mathcal{A}$ outputs the snapshot G_t's degree list $d_t = (d_t^+, d_t^-) \in [0, n-1]^n \times [0, n-1]^n$ while satisfying w-event local differential privacy.*

IV. Collection Algorithm

In this section, we present an algorithm for collecting degree lists from streaming directed graphs under w-event local differential privacy. Due to space constraints, some algorithms and proofs are provided in our technical report [42].

A. Overview

To implement w-event local differential privacy, we first show a novel composition theorem of event-level local differential privacy.

Theorem IV.1 (Composition of Event-Level Local Differential Privacy)**.** *Let $\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w$ be algorithms depending on different elements of input neighbor lists. If for every $i \in [w]$, $\mathcal{A}_i$ satisfies 1-event local differential privacy (i.e., event-level local differential privacy) with privacy budget ε_i, then the composition $(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w)$ satisfies w-event local differential privacy with privacy budget $\sum_{i=1}^{w} \varepsilon_i$.*

Different from basic composition. Let $\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w$ be algorithms depending on different elements of input neighbor lists. If for every $i \in [w]$, $\mathcal{A}_i$ satisfies 1-event local differential privacy with privacy budget ε_i, then by Theorem III.6, the basic composition $(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w)$ satisfies 1-event local differential privacy with privacy budget $\sum_{i=1}^{w} \varepsilon_i$.

By Theorem IV.1, we allocate a total privacy budget of ε to each user within any w-length interval. At each timestamp t, the algorithm releases the degree list of G_t using an event-level locally differentially private mechanism with budget ε/w per user. Theorem IV.1 then ensures that the resulting algorithm satisfies w-event local differential privacy with budget ε per user. At each timestamp, the algorithm consists of three phases: (1) neighbor list projection, (2) degree collection, and (3) post-processing.

Phase 1: Neighbor List Projection. At each timestamp, a user's incident out-edges and in-edges may undergo up to $n-1$ changes (insertions or deletions). Accordingly, the global sensitivity of each utility function is $n-1$, and each exponential mechanism has an output range of size n. These properties can limit utility. To address this issue, we apply neighbor list projection. After projection, each user's out-degree (resp. in-degree) is bounded by $\tilde{d}_{\max}^+$ (resp. $\tilde{d}_{\max}^-$). Consequently, the global sensitivities are reduced to $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$, respectively, and the corresponding output ranges contain $\tilde{d}_{\max}^+ + 1$ and $\tilde{d}_{\max}^- + 1$ values.

Phase 2: Degree Collection. Each user first computes their out-degree and in-degree, then uses the exponential mechanism to obtain estimates of both values. Each user reports these estimates to the central server, which aggregates them to produce an estimated degree list.

Phase 3: Post-Processing. The degree list output by **Phase 2** contains estimated in-degrees and out-degrees for all users, but it may not be digraphic; that is, no directed graph may realize it. In this phase, the central server applies Berger's necessary and sufficient characterization of digraphic degree lists [31] to adjust the estimates. The resulting degree list satisfies this characterization and is therefore digraphic.

B. Neighbor List Projection

For each user v_i at timestamp t, we use the exponential mechanism with privacy budget ε to estimate the out-degree under local differential privacy. We define the utility function for out-degree estimation as follows:

$$f_i^+ : \{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times [0, n-1] \to [-(n-1), 0],$$
$$(\mathbf{a}, r) \mapsto -|\sum_{j=1}^{n} a_j - r|.$$

We define the in-degree utility function similarly:

$$f_i^- : \{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times [0, n-1] \to [-(n-1), 0],$$
$$(\mathbf{a}, r) \mapsto -|\sum_{j=1}^{n} a_j - r|.$$

Because self-loops are not allowed in the input graph, both utility functions have domain $\{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times [0, n-1]$. The out-degree utility f_i^+ assigns higher utility to candidates $r \in [0, n-1]$ that are closer to user v_i's true out-degree, $\sum_{j=1}^n u_{i,t,j}^+$, in G_t. Similarly, f_i^- assigns higher utility to candidates closer to the true in-degree, $\sum_{i=1}^n u_{i,t,j}^-$. Since the exponential mechanism favors candidates with higher utility, these functions are suitable for estimating the out-degree and in-degree.

Proposition IV.2 (Global Sensitivity)**.** *For each user v_i, the global sensitivities of f_i^+ and f_i^- are both $n-1$.*

Proposition III.5 indicates that utility improves as the output range and global sensitivity decrease. We therefore project each user's out-neighbor list $\mathbf{u}_{i,t}^+$ to reduce both the candidate-set size $|\mathcal{R}|$ and the sensitivity $GS_{f_i^+}$. We apply the same projection to the in-neighbor list $\mathbf{u}_{i,t}^-$, reducing the corresponding output range and sensitivity for in-degree estimation.

Given thresholds $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$, we project each user's out-neighbor list as follows. If the list contains at most $\tilde{d}_{\max}^+$ entries equal to 1 (i.e., $\sum_{j=1}^n u_{i,t,j}^+ \leq \tilde{d}_{\max}^+$), we leave it unchanged. Otherwise, we randomly change enough entries from 1 to 0 so that the projected list contains exactly $\tilde{d}_{\max}^+$ entries equal to 1; that is, $\sum_{j=1}^n \tilde{u}_{i,t,j}^+ = \tilde{d}_{\max}^+$. We apply the same procedure to the in-neighbor list $\mathbf{u}_{i,t}^-$, ensuring that its projection $\tilde{\mathbf{u}}_{i,t}^-$ contains at most $\tilde{d}_{\max}^-$ entries equal to 1 (i.e., $\sum_{j=1}^n \tilde{u}_{i,t,j}^- \leq \tilde{d}_{\max}^-$). In practice, we need not explicitly construct the projected neighbor lists; it suffices to compute each user's projected out-degree and in-degree at each timestamp. For user v_i at timestamp t, these values are given by $\tilde{d}_{t,i}^+ \leftarrow \min\{d_{t,i}^+, \tilde{d}_{\max}^+\}$ and $\tilde{d}_{t,i}^- \leftarrow \min\{d_{t,i}^-, \tilde{d}_{\max}^-\}$. The projection bounds each user's out-degree and in-degree by $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$, respectively. Accordingly, the candidate ranges for the estimated out-degree and in-degree are $[0, \tilde{d}_{\max}^+]$ and $[0, \tilde{d}_{\max}^-]$, respectively. We therefore define the utility functions $\tilde{f}_i^+$ and $\tilde{f}_i^-$ over the respective domains $\{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times [0, \tilde{d}_{\max}^+]$ and $\{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times [0, \tilde{d}_{\max}^-]$.

Proposition IV.3. *For each user v_i, the global sensitivities of $\tilde{f}_i^+$ and $\tilde{f}_i^-$ are $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$, respectively.*

Example IV.4. *Consider a graph with vertex set $V = \{v_1, v_2, v_3, v_4, v_5\}$ and thresholds $\tilde{d}_{\max}^+ = 2$ and $\tilde{d}_{\max}^- = 1$. For v_3 at timestamp t, consider two neighbor lists $\mathbf{u}_{3,t} = (\boldsymbol{\lambda}, \boldsymbol{\mu}) = ((1,1,0,1,1),(1,1,0,1,1))$ and $\mathbf{u}'_{3,t} = (\boldsymbol{\lambda}', \boldsymbol{\mu}') = ((1,0,0,0,0),(1,0,0,0,1))$. The maximum differences between these lists, measured by the out-degree and in-degree utilities, are $\max_{r\in[0,4]} |f_3^+(\boldsymbol{\lambda}, r) - f_3^+(\boldsymbol{\lambda}', r)| = 3$ and $\max_{r\in[0,4]} |f_3^-(\boldsymbol{\mu}, r) - f_3^-(\boldsymbol{\mu}', r)| = 2$. Applying the projection rule gives $\sum_{j=1}^n \tilde{\lambda}_i = 2$, $\sum_{j=1}^n \tilde{\mu}_j = 1$, $\sum_{j=1}^n \tilde{\lambda}'_j = 1$, and $\sum_{j=1}^n \tilde{\mu}'_j = 1$. After projection, the corresponding maximum differences are $\max_{r\in[0,2]} |\tilde{f}_3^+(\boldsymbol{\lambda}, r) - \tilde{f}_3^+(\boldsymbol{\lambda}', r)| = 1 < \tilde{d}_{\max}^+$ and $\max_{r\in\{0,1\}} |\tilde{f}_3^-(\boldsymbol{\mu}, r) - \tilde{f}_3^-(\boldsymbol{\mu}', r)| = 0 < \tilde{d}_{\max}^-$.*

C. Degree Collection

In this phase, each user uses the exponential mechanism to locally estimate their out-degree and in-degree, with each estimate satisfying 1-event local differential privacy. Each user then reports both estimates to the central server, which aggregates them to form the degree list. The pseudocode and a detailed complexity analysis appear in our technical report [42].

Complexity. The per-user running time is $O(\tilde{d}_{\max}^+ + \tilde{d}_{\max}^-)$, while the central server's running time is $O(n)$. The communication complexity is $O(n)$.

D. Post-Processing

The degree list $\hat{d}_t$ produced in **Phase 2** may not be digraphic, meaning that no directed graph realizes it. To address this, the central server applies a post-processing step after aggregating the users' degree reports, producing a digraphic degree list.

To make the degree list digraphic, we use the discrete mathematics result of [31],

Theorem IV.5. *Let $d = ((d_1^+, d_2^+, \cdots, d_n^+), (d_1^-, d_2^-, \cdots, d_n^-)) \in \mathbb{Z}_{\geq 0}^n \times \mathbb{Z}_{\geq 0}^n$ be such that $d_1^- \geq d_2^- \geq \cdots \geq d_n^-$. Then, d is digraphic if and only if,*

1) $\sum_{i=1}^n d_i^+ = \sum_{i=1}^n d_i^-$,
2) $\forall k \in \{q \in [n-1] : d_q^- > d_{q+1}^-\} \cup \{n\}$, $\sum_{i=1}^k \min\{d_i^+, k-1\} + \sum_{i=k+1}^n \min\{d_i^+, k\} \geq \sum_{i=1}^k d_i^-$.

Theorem IV.5 states a necessary and sufficient condition, thus we only need to make the degree list satisfy this condition, then the degree list will be digraphic. Observing this condition, it has two constraint. Similar to the handshaking lemma in undirected graphs, the first constraint is trivial. Therefore, the post-processing algorithm first makes the degree list satisfy the second constraint. Then, the algorithm makes the degree list satisfy the first constraint while not breaking the second constraint.

Theorem IV.6. *Let $n \in \mathbb{N}$, and let $(\hat{d}_t^+, \hat{d}_t^-) \in \mathbb{Z}_{\geq 0}^n \times \mathbb{Z}_{\geq 0}^n$. Then, the output of Algorithm 1 is a digraphic non-negative integer list.*

Proof. The algorithm first sorts the $(\hat{d}_t^+, \hat{d}_t^-)$ and obtains a degree list $(\overline{d}_t^+, \overline{d}_t^-) \in \mathbb{Z}_{\geq 0}^n \times \mathbb{Z}_{\geq 0}^n$ such that $\overline{d}_{t,1}^- \geq \overline{d}_{t,2}^- \geq \cdots \geq \overline{d}_{t,n}^-$.

After sorting, the algorithm loops k from $n-1$ to 1. For all k such that $\sum_{i=1}^k \min\{\overline{d}_{t,i}^+, k-1\} + \sum_{i=k+1}^n \min\{\overline{d}_{t,i}^+, k\} < \sum_{i=1}^k \overline{d}_{t,i}^-$, if $\overline{d}_{t,k}^- > \overline{d}_{t,k+1}^-$, then the algorithm sets $\overline{d}_{t,k}^- = \overline{d}_{t,k+1}^-$. Therefore, $(\overline{d}_t^+, \overline{d}_t^-)$ satisfies 2) in Theorem IV.5 (except the case that $k = n$).

Next, the algorithm makes $(\overline{d}_t^+, \overline{d}_t^-)$ satisfy 1) in Theorem IV.5, which implies the case $k = n$ of 2) in Theorem IV.5. There are two cases: (i) $\sum_{i=1}^n \overline{d}_{t,i}^+ > \sum_{i=1}^n \overline{d}_{t,i}^-$; (ii) $\sum_{i=1}^n \overline{d}_{t,i}^+ < \sum_{i=1}^n \overline{d}_{t,i}^-$.

(i) **Case** $\sum_{i=1}^n \overline{d}_{t,i}^+ > \sum_{i=1}^n \overline{d}_{t,i}^-$: The queue Q contains all $k \in [n-1]$ such that $\overline{d}_{t,k}^- > \overline{d}_{t,k+1}^-$. The order of elements in Q is descending. For every $i \in [k]$ such that $\overline{d}_{t,i}^+ > k-1$, if $\sum_{i=1}^n \overline{d}_{t,i}^+ > \sum_{i=1}^n \overline{d}_{t,i}^-$ still holds, then the algorithm reduces $\overline{d}_{t,i}^+$. For every $i \in [k+1, n]$ such

Algorithm 1: Post-Processing

Input : The estimated degree list $\hat{d}_t = (\hat{d}_t^+, \hat{d}_t^-)$; the number of users n
Output: The post-processing degree list $\overline{d}_t = (\overline{d}_t^+, \overline{d}_t^-)$

1 Find $\sigma \in S_n$, s.t. $\hat{d}^-_{t,\sigma(1)} \geq \hat{d}^-_{t,\sigma(2)} \geq \cdots \geq \hat{d}^-_{t,\sigma(n)}$;
2 $\overline{d}_t \leftarrow ((\hat{d}^+_{t,\sigma(1)}, \cdots, \hat{d}^+_{t,\sigma(n)}), (\hat{d}^-_{t,\sigma(1)}, \cdots, \hat{d}^-_{t,\sigma(n)}))$;
3 Queue $Q \leftarrow \emptyset$;
4 **for** $k = n-1$ *to* 1 **do**
5 **if** $\overline{d}^-_{t,k} == \overline{d}^-_{t,k+1}$ **then**
6 **continue**;
7 $l \leftarrow \sum_{i=1}^{k} \min\{\overline{d}^+_{t,i}, k-1\} + \sum_{i=k+1}^{n} \min\{\overline{d}^+_{t,i}, k\}$;
8 $r \leftarrow \sum_{i=1}^{k} \overline{d}^-_{t,i}$;
9 **if** $l < r$ **then**
10 $\overline{d}^-_{t,k} \leftarrow \overline{d}^-_{t,k+1}$;
11 **else**
12 $Q \leftarrow Q \cup \{k\}$;
13 $\overline{d}^+_{t,\text{sum}} \leftarrow \sum_{i=1}^{n} \overline{d}^+_{t,i}$, $\overline{d}^-_{t,\text{sum}} \leftarrow \sum_{i=1}^{n} \overline{d}^-_{t,i}$;
14 **if** $\overline{d}^+_{t,\text{sum}} > \overline{d}^-_{t,\text{sum}}$ **then**
15 **while** $Q \neq \emptyset$ **do**
16 Pop k from Q;
17 **for** $i = 1$ *to* n **do**
18 $\tau \leftarrow k - \mathbb{I}(i \leq k)$;
19 **if** $\overline{d}^+_{t,i} > \tau$ **then**
20 $\Delta \leftarrow \overline{d}^+_{t,i} - \tau$;
21 **if** $\Delta \geq \overline{d}^+_{t,\text{sum}} - \overline{d}^-_{t,\text{sum}}$ **then**
22 $\overline{d}^+_{t,i} \leftarrow \overline{d}^+_{t,i} - (\overline{d}^+_{t,\text{sum}} - \overline{d}^-_{t,\text{sum}})$, $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^-_{t,\text{sum}}$;
23 **break while**;
24 **else**
25 $\overline{d}^+_{t,i} \leftarrow \tau$, $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^+_{t,\text{sum}} - \Delta$;
26 **if** $\overline{d}^+_{t,\text{sum}} > \overline{d}^-_{t,\text{sum}}$ **then**
27 **for** $i = 1$ *to* n **do**
28 **if** $\overline{d}^+_{t,i} \geq \overline{d}^+_{t,\text{sum}} - \overline{d}^-_{t,\text{sum}}$ **then**
29 $\overline{d}^+_{t,i} \leftarrow \overline{d}^+_{t,i} - (\overline{d}^+_{t,\text{sum}} - \overline{d}^-_{t,\text{sum}})$, $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^-_{t,\text{sum}}$;
30 **break**;
31 **else**
32 $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^+_{t,\text{sum}} - \overline{d}^+_{t,i}$, $\overline{d}^+_{t,i} \leftarrow 0$;
33 **else if** $\overline{d}^+_{t,\text{sum}} < \overline{d}^-_{t,\text{sum}}$ **then**
34 **for** $i = 1$ *to* n **do**
35 $\Delta \leftarrow (n-1) - \overline{d}^+_{t,i}$;
36 **if** $\Delta \geq \overline{d}^-_{t,\text{sum}} - \overline{d}^+_{t,\text{sum}}$ **then**
37 $\overline{d}^+_{t,i} \leftarrow \overline{d}^+_{t,i} + (\overline{d}^-_{t,\text{sum}} - \overline{d}^+_{t,\text{sum}})$, $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^-_{t,\text{sum}}$;
38 **break**;
39 **else**
40 $\overline{d}^+_{t,i} \leftarrow n-1$, $\overline{d}^+_{t,\text{sum}} \leftarrow \overline{d}^+_{t,\text{sum}} + \Delta$;
41 $\overline{d}_t \leftarrow ((\overline{d}^+_{t,\sigma^{-1}(1)}, \cdots, \overline{d}^+_{t,\sigma^{-1}(n)}), (\overline{d}^-_{t,\sigma^{-1}(1)}, \cdots, \overline{d}^-_{t,\sigma^{-1}(n)}))$;
42 **return** $\overline{d}_t$;

that $\overline{d}^+_{t,i} > k$, the algorithm makes the same operation. Since $\overline{d}^+_{t,i}$ at most reduced to $k-1$ (if $i \in [k]$) or k (if $i \in [k+1, n]$), $(\overline{d}_t^+, \overline{d}_t^-)$ still satisfies $\sum_{i=1}^{k} \min\{\overline{d}^+_{t,i}, k-1\} + \sum_{i=k+1}^{n} \min\{\overline{d}^+_{t,i}, k\} \geq \sum_{i=1}^{k} \overline{d}^-_{t,i}$. Next, we show that for any smaller or larger $k' \in [n-1]$ such that $\overline{d}^-_{t,k'} > k'-1$, $\sum_{i=1}^{k'} \min\{\overline{d}^+_{t,i}, k'-1\} + \sum_{i=k'+1}^{n} \min\{\overline{d}^+_{t,i}, k'\} \geq \sum_{i=1}^{k'} \overline{d}^-_{t,i}$ still holds while reducing these elements.

- For a fixed k' such that $0 \leq k' < k$ and $\overline{d}^-_{t,k'} > \overline{d}^-_{t,k'+1}$. For any $i \in [k]$ such that $\overline{d}^+_{t,i} \geq k-1$, we have $\overline{d}^+_{t,i} \geq k'$. For any $i \in [k+1, n]$ such that $\overline{d}^+_{t,i} \geq k$, we have $\overline{d}^+_{t,i} \geq k'$. Therefore, after reducing, $\overline{d}^-_{t,k'} > k'-1$, $\sum_{i=1}^{k'} \min\{\overline{d}^+_{t,i}, k'-1\} + \sum_{i=k'+1}^{n} \min\{\overline{d}^+_{t,i}, k'\} \geq \sum_{i=1}^{k'} \overline{d}^-_{t,i}$ still holds.
- For a fixed k' such that $k < k' \leq n-1$ and $\overline{d}^-_{t,k'} > \overline{d}^-_{t,k'+1}$. The algorithm will not pop k from Q, unless for every $i \in [k']$, $\overline{d}^-_{t,i} = k'-1$ and for every $i \in [k'+1, n]$, $\overline{d}^-_{t,i} = k'$. In this case, notice that for any $i \in [n]$, the algorithm may only reduce $\overline{d}^-_{t,i}$. Therefore, 1) in Theorem IV.5 implies $\sum_{i=1}^{k'} \min\{\overline{d}^+_{t,i}, k'-1\} + \sum_{i=k'+1}^{n} \min\{\overline{d}^+_{t,i}, k'\} \geq \sum_{i=1}^{k'} \overline{d}^-_{t,i}$.

If the algorithm has traversed all $k \in [n-1]$ such that $\overline{d}^-_{t,k} > \overline{d}^-_{t,k+1}$ and it still holds $\sum_{i=1}^{n} \overline{d}^+_{t,i} > \sum_{i=1}^{n} \overline{d}^-_{t,i}$, then the algorithm loops i from 1 to n. For any $i \in [n]$, if $\sum_{i=1}^{n} \overline{d}^+_{t,i} > \sum_{i=1}^{n} \overline{d}^-_{t,i}$ still holds, then the algorithm reduces $\overline{d}^+_{t,i}$. Fix a $k \in [n-1]$ such that $\overline{d}^-_{t,k} > \overline{d}^-_{t,k+1}$. For any $i \in [k]$, we have $\overline{d}^-_{t,k} \leq k-1$. For any $i \in [k+1, n]$, we have $\overline{d}^-_{t,k} \leq k$. Therefore, 1) in Theorem IV.5 implies $\sum_{i=1}^{k} \min\{\overline{d}^+_{t,i}, k-1\} + \sum_{i=k+1}^{n} \min\{\overline{d}^+_{t,i}, k\} \geq \sum_{i=1}^{k} \overline{d}^-_{t,i}$.
(ii) **Case** $\sum_{i=1}^{n} \overline{d}^+_{t,i} < \sum_{i=1}^{n} \overline{d}^-_{t,i}$: The algorithm loops i from 1 to n. For any $i \in [n]$, if $\overline{d}^+_{t,i} < \overline{d}^-_{t,i}$ still holds, then the algorithm increases $\overline{d}^+_{t,i}$.

Therefore, $(\overline{d}_t^+, \overline{d}_t^-)$ is digraphic. Finally, the algorithm restores the original vertex order. After restoring, $(\overline{d}_t^+, \overline{d}_t^-)$ is still digraphic. □

As shown in Algorithm 1, the algorithm first sorts the vertices in non-increasing order of their in-degrees using a permutation σ (Line 1). It then reorders the degree list according to σ and initializes an empty queue Q (Lines 2, 3). For each k from $n-1$ down to 1, the algorithm skips the index if the in-degrees at positions k and $k+1$ are equal. Otherwise, it evaluates the two sides, l and r, of the corresponding inequality. If $l < r$, it lowers the in-degree at position k to match the one at position $k+1$; otherwise, it adds k to Q (Lines 4-12). After this scan, the algorithm computes the total out-degree and in-degree (Line 13). If the total out-degree is larger, it processes the indices in Q, scanning the vertices and capping each out-degree at k for indices $i \leq k$ and at $k-1$ for indices $i > k$ (Lines 14-25). If an excess remains after the queue is exhausted, it reduces the out-degrees sequentially until the totals match (Lines 26-32). If the total out-degree is smaller, it increases the out-degrees sequentially, up to $n-1$ per vertex, until the totals match (Lines 33-40). Finally, it applies σ^{-1} to restore the original vertex order and returns the adjusted degree list (Lines 41, 42).

Complexity. We now analyze the running time of the post-processing phase. Sorting the estimated degree list $\hat{d}_t$ in non-increasing order of in-degree takes $O(n \log n)$ time. The algorithm then scans the indices k, which takes $O(n)$ time. The body of this scan is executed only when $k \in [n-1]$ satisfies $\overline{d}^-_{t,k} > \overline{d}^-_{t,k+1}$. Since the in-degrees are bounded by $\tilde{d}^-_{\max}$, this condition holds for at most $\tilde{d}^-_{\max} - 1$ indices. Each iteration takes $O(n)$ time, so the scan takes $O(n\tilde{d}^-_{\max})$ time in total. The

Algorithm 2: Online Degree List Collection

Input : Streaming directed graph $\mathbf{G}$ represented as $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n$; the thresholds $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$; the number of users n; window length w; privacy budget $\varepsilon \in \mathbb{R}_{\geq 0}$

Output: Released degree list $\overline{d}_t = (\overline{d}^+_t, \overline{d}^-_t)$ at each timestamp t

1 $t \leftarrow 1$;
2 **while** G_t *exists* **do**
3 Estimated Degree list $\hat{d}_t \leftarrow$Degree Collection(G_t, $\tilde{d}^+_{\max}$, $\tilde{d}^-_{\max}$, n , ε/w);
4 Post-processing degree list $\overline{d}_t \leftarrow$Post-Processing($\hat{d}_t$, n);
5 **Output** $\overline{d}_t$;
6 $t \leftarrow t + 1$;

subsequent adjustment of the out-degrees also takes at most $O(n\tilde{d}^-_{\max})$ time, because $|Q| \leq \tilde{d}^-_{\max} - 1$ and processing each queue entry requires at most $O(n)$ time. Restoring the original vertex order using the inverse permutation σ^{-1} takes $O(n)$ time. Therefore, the total running time of the post-processing algorithm at the central server is $O(n \log n + n\tilde{d}^-_{\max})$. Here, $\tilde{d}^-_{\max}$ denotes the projection threshold for the in-neighbor lists on each user side.

E. Overall Algorithm

Given a finite or infinite streaming directed graph $\mathbf{G}$, represented by neighbor lists $\mathbf{u}_1, \ldots, \mathbf{u}_n$, projection thresholds $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$, window length w, and privacy budget ε, we call the proposed method the Online Degree List Collection (ODLC) algorithm. It initializes $t = 1$ and processes each available snapshot G_t in sequence. At each timestamp, it uses a per-timestamp privacy budget of ε/w, obtains an estimated degree list $\hat{d}_t$ through degree collection (Line 3), applies post-processing to produce an adjusted degree list $\overline{d}_t$ (Line 4), and releases the result (Line 5). It then advances to the next timestamp and repeats; for an infinite stream, it continues processing snapshots online.

Theorem IV.7. *Algorithm 2 satisfies w-event local differential privacy for each user with privacy budget ε.*

Complexity. At each timestamp t, ODLC takes $O(\tilde{d}^+_{\max} + \tilde{d}^-_{\max})$ time on each user side and $O(n \log n + n\tilde{d}^-_{\max})$ time on the central server. Its communication cost is $O(n)$ per timestamp.

V. Optimized Algorithm

In this section, we present Optimized Online Degree List Collection (O-ODLC) algorithm, an optimized variant of ODLC algorithm for degree list collection in streaming directed graphs under w-event local differential privacy.

A. Beyond Sparsity

Let $\mathbb{P}$ be a discrete probability measure on a discrete sample space Ω. When the cardinality of Ω is large, most individual outcomes may have very small probabilities.

Proposition V.1. *Let Ω be a discrete sample space. Let $\mathbb{P}$ be a discrete probability measure on Ω. For any $\delta \in (0, 1)$, define $A_\delta := \{x \in \Omega : \mathbb{P}(\{x\}) > \delta\}$. Then, we have $|A_\delta| < \frac{1}{\delta}$.*

Proposition V.1 implies that, for a fixed probability threshold δ, the number of outcomes with probability greater than δ is strictly less than $\frac{1}{\delta}$. This shows that when Ω is sufficiently large, outcomes with probability at most δ become the majority. We refer to this phenomenon as sparsity. For example, let $n \in \mathbb{N}$, and assume $\mathbb{P}$ is uniform on $[n]$. Then, for any $i \in [n]$, we have $\mathbb{P}(\{i\}) = \frac{1}{n}$, which is very small when n is large. When n is equal to 10000, it is 0.0001. Notice that, in the exponential mechanism, if the output range $\mathcal{R}$ contains too many elements, then the output distribution of the algorithm may become sparsity. The sparsity phenomenon may reduce the utility of the algorithmic output, since most of elements in the output range are selected with approximately equal probability.

In fact, in ODLC algorithm, the output ranges of the exponential mechanisms are $[0, \tilde{d}^+_{\max}]$ and $[0, \tilde{d}^-_{\max}]$, respectively. Here, $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$ are the projection thresholds. Although $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$ are both much less than the number of vertices, the output ranges $[0, \tilde{d}^+_{\max}]$ and $[0, \tilde{d}^-_{\max}]$ are still large. To beyond the sparsity phenomenon, we use a step size θ. For a fixed $\theta \in \mathbb{N}$, we set the output ranges of the exponential mechanisms as $\{j\theta : 0 \leq j \leq \tilde{d}^+_{\max}/\theta, j \in \mathbb{Z}\}$ and $\{j\theta : 0 \leq j \leq \tilde{d}^-_{\max}/\theta, j \in \mathbb{Z}\}$, respectively. Thus, the output ranges are reduced, thereby mitigating the sparsity phenomenon. The utility of the algorithmic output is improved.

Notice that for each user v_i, the domain of the utility functions $\tilde{f}^+_i$ and $\tilde{f}^-_i$ have been changed, since we introduce the step size θ in the algorithm. For each user v_i, we consider the new utility functions $\tilde{f}^+_{i,\theta}$ and $\tilde{f}^-_{i,\theta}$. The domain of $\tilde{f}^+_{i,\theta}$ is $\{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times \{j\theta : 0 \leq j \leq \tilde{d}^+_{\max}/\theta, j \in \mathbb{Z}\}$. Similarly, the domain of $\tilde{f}^-_{i,\theta}$ is $\{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\} \times \{j\theta : 0 \leq j \leq \tilde{d}^-_{\max}/\theta, j \in \mathbb{Z}\}$. Therefore, to use the exponential mechanism correctly, we calculate the global sensitivity of the $\tilde{f}^+_{i,\theta}$ and $\tilde{f}^-_{i,\theta}$, respectively.

Proposition V.2. *For any $\theta \in \mathbb{N}$ and each user v_i, the global sensitivities of $\tilde{f}^+_{i,\theta}$ and $\tilde{f}^-_{i,\theta}$ are $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$, respectively.*

B. Efficiency Optimization

Snapshots of a streaming directed graph often share many edges across timestamps: some edges are updated at each timestamp, while many remain unchanged. Consequently, each vertex's in-degree and out-degree tend to remain relatively stable over time. Moreover, these degrees are also generally small but nonzero. Their estimates need not be refreshed at every timestamp. At each timestamp, the exponential mechanism need only be applied to significant degrees, thereby reducing its computational cost.

Inspired by the sparse vector technique [38], [43], [44], at each timestamp, each user determines whether to update the estimated out-degree and in-degree. The update decision (denoted by $\top$) and the decision to retain the current estimate (denoted by $\perp$) must also satisfy w-event local differential privacy; otherwise, the decision itself may reveal private information. To demonstrate why this matters, we show an incorrect approach and explain how privacy leaks can occur: each user v_i applies two fixed thresholds, η_1 and η_2, satisfying

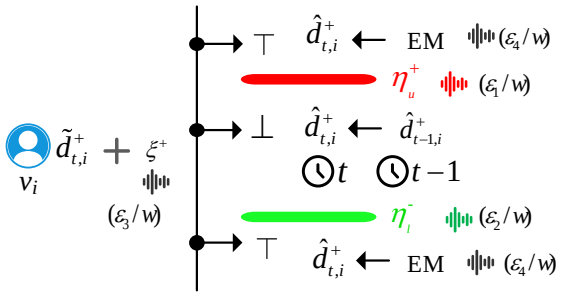


Fig. 4: The determination process workflow

$\eta_1, \eta_2 \in \mathbb{Z}_{\geq 0}$ and $\eta_1 < \eta_2$. At timestamp t, v_i then determines that the estimated out-degree (resp. estimated in-degree) should be updated if and only if $d^+_{t,i} \leq \eta_1$ or $d^+_{t,i} \geq \eta_2$ (resp. $d^-_{t,i} \leq \eta_1$ or $d^-_{t,i} \geq \eta_2$). It is easy to verify that this rule does not satisfy w-event local differential privacy. We next show that using this $\top/\perp$ rule in the degree list collection algorithm of Section IV violates w-event local differential privacy for user v_i. Let $\mathcal{A}_{\text{error}}$ denote the resulting collection algorithm. Consider two possible neighbor lists, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, for user v_i such that $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are w-neighboring and satisfy:

1) $\forall t \in \mathcal{I}$, $d^+(t, \boldsymbol{\alpha}) \geq \eta_2$,
2) $\forall t \in \mathcal{I}$, $d^-(t, \boldsymbol{\alpha}) \geq \eta_2$,
3) $\forall t \in \mathcal{I}$, $\eta_1 < d^+(t, \boldsymbol{\beta}) < \eta_2$,
4) $\forall t \in \mathcal{I}$, $\eta_1 < d^-(t, \boldsymbol{\beta}) < \eta_2$.

Here, $\mathcal{I} = \{t : \boldsymbol{\alpha}_t \neq \boldsymbol{\beta}_t\}$. d^+ and d^- denote the out-degree function and in-degree function, respectively. Given a neighbor list $\mathbf{u}_i$ and timestamp t, d^+ (resp. d^-) maps $(\mathbf{u}_i, t)$ to the out-degree (resp. in-degree) of v_i at timestamp t. Let $l = \min \mathcal{I}$ and define the following output event for $\mathcal{A}_{\text{error}}$: $\mathcal{O} = \{o \in \text{Range}(\mathcal{A}_{\text{error}}) : \exists t \in \mathcal{I}, o^+_{t,i} \neq o^+_{l-1,i} \vee o^-_{t,i} \neq o^-_{l-1,i}\}$ (o_t denote the released degree list $\overline{d}_t$ by $\mathcal{A}_{\text{error}}$ at timestamp t; we set $o_{0,i} = 0$ in $\mathcal{A}_{\text{error}}$) , then

$$\Pr[\mathcal{A}_{\text{error}}(\boldsymbol{\alpha}) \in \mathcal{O}] = 0 \quad \text{and} \quad \Pr[\mathcal{A}_{\text{error}}(\boldsymbol{\beta}) \in \mathcal{O}] > 0.$$

By the privacy condition in Definition III.2, if $\mathcal{A}_{\text{error}}$ satisfied w-event local differential privacy for user v_i, it would have to hold that

$$\Pr[\mathcal{A}_{\text{error}}(\boldsymbol{\beta}) \in \mathcal{O}] \leq e^{\varepsilon} \Pr[\mathcal{A}_{\text{error}}(\boldsymbol{\alpha}) \in \mathcal{O}] = 0.$$

This is a contradiction. Hence, $\mathcal{A}_{\text{error}}$ does not satisfy w-event local differential privacy for user v_i.

To ensure that the optimized collection algorithm satisfies w-event local differential privacy for each user, we allocate part of the privacy budget to the update decisions for the estimated in-degree and out-degree. At each timestamp t, for each user v_i, we determine in turn whether to update the estimated out-degree and the estimated in-degree. To decide whether to update the estimated out-degree, we use three random variables, η^+_u, η^+_l, and ξ^+, which follow Laplace distributions with location parameter 0 and scale parameters $\frac{w\tilde{d}^+_{\max}}{\varepsilon_1}$, $\frac{w\tilde{d}^+_{\max}}{\varepsilon_2}$, and $\frac{2w\tilde{d}^+_{\max}}{\varepsilon_3}$, respectively. Here, w is the window length, $\tilde{d}^+_{\max}$ is the projection threshold used in the neighbor list projection, and $\varepsilon_1, \varepsilon_2, \varepsilon_3$ are three separate privacy budgets. Fig. 4 illustrates how the algorithm decides whether to update $\hat{d}^+_{t,i}$ using the exponential mechanism or retain its previous value $\hat{d}^+_{t-1,i}$. After the neighbor list projection, the user has the out-degree $\tilde{d}^+_{t,i}$ (i.e., $\min\{d^+_{t,i}, \tilde{d}^+_{\max}\}$). The algorithm first generates random thresholds η^+_u and η^+_l, respectively. The corresponding privacy budgets are $\frac{\varepsilon_1}{w}$ and $\frac{\varepsilon_2}{w}$, respectively. The algorithm then samples Laplace noise ξ^+ and adds it to the out-degree $\tilde{d}^+_{t,i}$, consuming the privacy budget $\frac{\varepsilon_3}{w}$. The algorithm then compares $\tilde{d}^+_{t,i} + \xi^+$ with the thresholds η^+_u and η^+_l. If $\tilde{d}^+_{t,i} + \xi^+ \geq \eta^+_u$ or $\tilde{d}^+_{t,i} + \xi^+ \leq \eta^-_l$, the algorithm applies the exponential mechanism to update the estimated out-degree $\hat{d}^+_{t,i}$, which consumes the privacy budget $\frac{\varepsilon_3}{w}$. If $\eta^+_l < \tilde{d}^+_{t,i} < \eta^+_u$, the algorithm retains the previous out-degree estimate by setting $\hat{d}^+_{t,i}$ to $\hat{d}^+_{t-1,i}$. The in-degree estimate $\hat{d}^-_{t,i}$ is updated via the exponential mechanism or retained as $\hat{d}^-_{t-1,i}$ according to the same rule.

Difference from the Sparse Vector Technique. The sparse vector technique allocates more of its privacy budget to significant queries, with the goal of improving utility. In contrast, our optimization uses a small portion of the privacy budget to identify significant degrees and applies the exponential mechanism only to those degrees. As a result, it reduces the algorithm's running time.

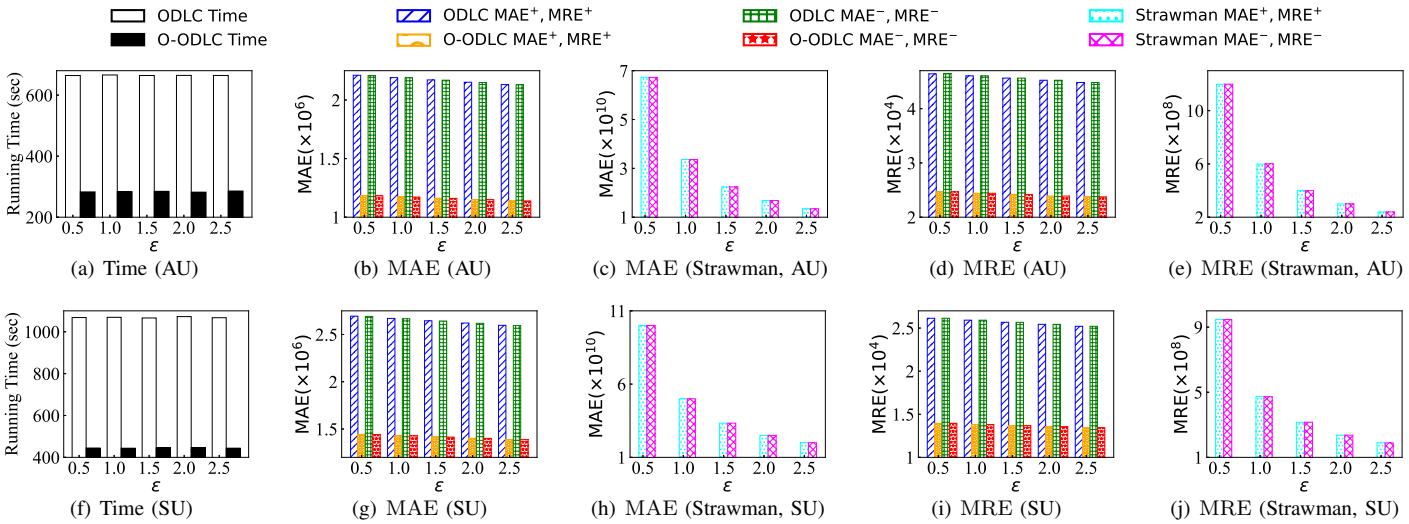
As shown in Algorithm 3, the algorithm initializes the time step t to 1 (Line 1). Then, for each snapshot G_t, processing proceeds on both the user side and the central-server side. On each user v_i's side, for each sign $s \in \{+, -\}$

Algorithm 3: Optimized Online Degree List Collection

Input : Streaming directed graph $\mathbf{G}$ represented as $\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n$; the thresholds $\tilde{d}^+_{\max}$ and $\tilde{d}^-_{\max}$; the number of users n; window length w; the step size θ; privacy budget $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \mathbb{R}_{>0}, \varepsilon_4 \in \mathbb{R}_{\geq 0}$

Output: Released degree list $\overline{d}_t = (\overline{d}^+_t, \overline{d}^-_t)$ at each timestamp t

1 $t \leftarrow 1$;
2 **while** G_t *exists* **do**
// On each user side
3 **for** $i = 1$ *to* n **do**
4 **for** *sign* s *in* $\{+, -\}$ **do**
5 $\eta^s_u \leftarrow \text{Lap}(\frac{w\tilde{d}^s_{\max}}{\varepsilon_1})$, $\eta^s_l \leftarrow \text{Lap}(\frac{w\tilde{d}^s_{\max}}{\varepsilon_2})$;
6 $\xi^s \leftarrow \text{Lap}(\frac{2w\tilde{d}^s_{\max}}{\varepsilon_3})$;
7 $\chi^s \leftarrow \min\{d^s_{t,i}, \tilde{d}^s_{\max}\} + \xi^s$;
8 **if** $\chi^s \leq \eta^s_l$ *or* $\chi^s \geq \eta^s_u$ **then**
9 Let $c \leftarrow 0$;
10 **for** $k = 0$ *to* $\tilde{d}^s_{\max}$ **step** θ **do**
11 $p^s(k) \leftarrow \exp(-\frac{\varepsilon_4 |\min\{d^s_{t,i}, \tilde{d}^s_{\max}\} - k|}{2w\tilde{d}^s_{\max}})$;
12 $c \leftarrow c + 1$;
13 $p^s_{\text{sum}} \leftarrow \sum_{j=0}^{c-1} p^s(j\theta)$;
14 $\mathbf{p} \leftarrow (\frac{1}{p^s_{\text{sum}}}(p^s(0), p^s(\theta), \cdots, p^s((c-1)\theta)))$;
15 $\hat{d}^+_{t,i} \leftarrow \text{Categorical}(\mathbf{p})$;
16 Report $\hat{d}^s_{t,i}$ to the central server;
17 **else**
// Define $\hat{d}^s_{t,0} = 0$
18 $\hat{d}^s_{t,i} \leftarrow \hat{d}^s_{t-1,i}$;
19 Report $\hat{d}^s_{t,i}$ to the central server;
// On the central server side
20 Aggregate $\hat{d}^+_t = (\hat{d}^+_{t,1}, \hat{d}^+_{t,2}, \cdots, \hat{d}^+_{t,n})$;
21 Aggregate $\hat{d}^-_t = (\hat{d}^-_{t,1}, \hat{d}^-_{t,2}, \cdots, \hat{d}^-_{t,n})$;
22 Post-processing degree list $\overline{d}_t \leftarrow$Post-Processing($\hat{d}_t$, n);
23 **Output** $\overline{d}_t$;
24 $t \leftarrow t + 1$;



(a) Time (AU) (b) MAE (AU) (c) MAE (Strawman, AU) (d) MRE (AU) (e) MRE (Strawman, AU)

(f) Time (SU) (g) MAE (SU) (h) MAE (Strawman, SU) (i) MRE (SU) (j) MRE (Strawman, SU)

Fig. 5: Impact of privacy budget ε

(s distinguishes the out-degree and in-degree components, respectively), the algorithm first determines the values of the random variables η_u^s, η_l^s, and ξ^s by sampling, with scales based on the window length w, the threshold $\tilde{d}_{\max}^s$, and the separate privacy budgets ε_1, ε_2, ε_3 (Lines 5, 6). Then, it computes the noisy degree χ^s (Line 7). Next, the algorithm checks whether χ^s satisfies $\chi^s \leq \eta_l^s$ or $\chi^s \geq \eta_u^s$ (Line 8). If so, it prepares a categorical distribution by initializing c as 0 (Line 9), and for $j \in \{j : 0 \leq j \leq \tilde{d}_{\max}^s/\theta, j \in \mathbb{Z}\}$, it computes the weight $p^s(j\theta)$ and updates c (Lines 11, 12). After computing the total weight p_{sum}^s (Line 13), the algorithm samples the estimated degree $\hat{d}_{t,i}^s$ from the categorical distribution generated by all the weights and reports it to the central server (Lines 14-16). Otherwise, if $\eta_l^s < \chi^s < \eta_u^s$, the algorithm sets the estimated degree $\hat{d}_{t,i}^s$ as the previous estimated degree $\hat{d}_{t-1,i}^s$ (Lines 18, 19). On the central server side, the algorithm aggregates all reported out-degrees (resp. in-degrees) to aggregate the estimated out-degree (resp. in-degree) lists $\hat{d}_t^+$ (resp. $\hat{d}_t^-$) (Lines 20, 21). Then, it applies the post-processing algorithm (i.e., Algorithm 1) to adjust the degree list, resulting in $\overline{d}_t$ (Line 22). Finally, the algorithm outputs $\overline{d}_t$ and increments the timestamps t (Lines 23, 24).

Theorem V.3. *Algorithm 3 satisfies w-event local differential privacy for each user with privacy budget $(\varepsilon_1+\varepsilon_2+\varepsilon_3+\varepsilon_4)$.*

Complexity. At each timestamp t, Algorithm 3 takes $O((\tilde{d}_{\max}^+ + \tilde{d}_{\max}^-)/\theta)$ time on each user side; takes $O(n \log n + n\tilde{d}_{\max}^-)$ time on the central server side; the communication complexity is $O(n)$.

VI. Experiments

We conduct our experiments on a server equipped with an Intel(R) Xeon(R) CPU E5-2650 processor and 128 GB of main memory. All experiments are implemented in C++ and run on CentOS. Due to space limitations, some experimental results can be found in our technical report [42].

A. Experimental Settings

Strawman Solution. SPAS [36] is a state-of-the-art method for continuously releasing data streams under w-event differential privacy, and improves utility through privacy budget allocation. We adapt the SPAS framework to Problem III.10 as our strawman solution. This solution does not include the neighbor list projection or post-processing phases.

TABLE I: Statistics of the datasets

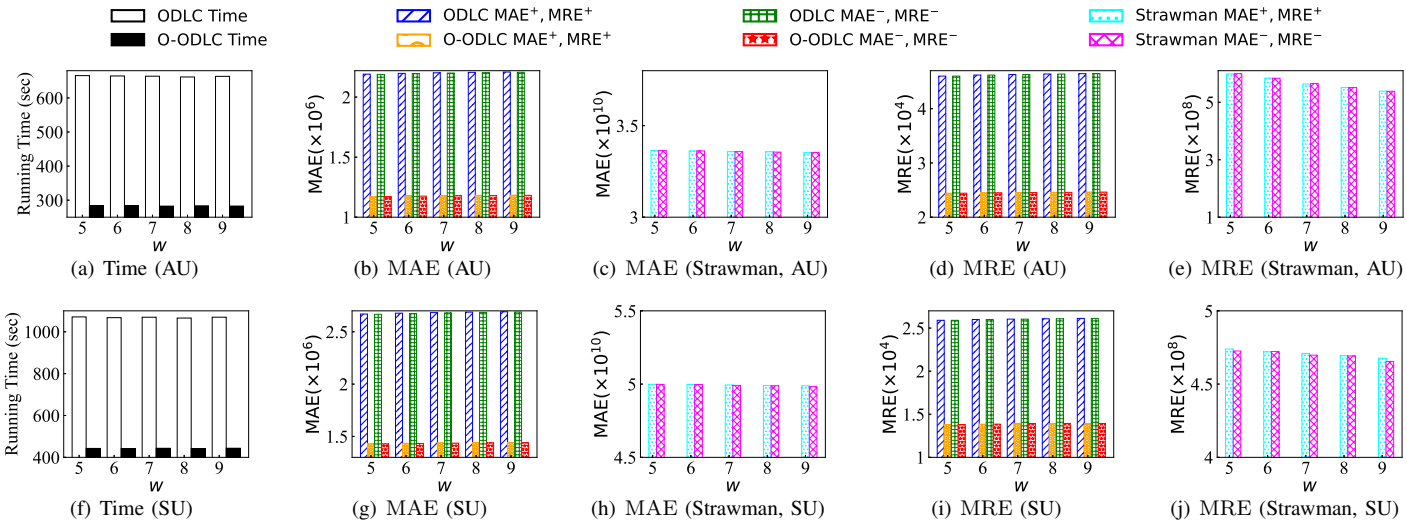
Dataset	$\lvert V\rvert$	$\lvert E\rvert$	T	$\deg_{\max}^+$	$\deg_{\max}^-$
Math Overflow (MO)	24.8**K**	0.9**K**	336	21.4881	21.0357
Ask Ubuntu (AU)	159.3**K**	1.9**K**	302	75.4205	27.6523
Super User (SU)	194.1**K**	2.4**K**	388	80.5438	27.1134
Wiki-Talk (WT)	1.1**M**	12.4**K**	317	1119.83	43.429

Datasets. TABLE I summarizes the datasets, all of which are from SNAP [45]. Each snapshot contains the edges observed over one week. Here, $|V|$ is the number of vertices, $|E|$ is the average number of edges per timestamp, and T is the number of timestamps. The quantities $\deg_{\max}^+$ and $\deg_{\max}^-$ denote, respectively, the average over timestamps of the maximum out-degree and maximum in-degree among all vertices.

Algorithms. We evaluate the **ODLC** algorithm from Section IV and the **O-ODLC** algorithm from Section V, and compare both with the **strawman** solution. Unless otherwise stated, we set the privacy budget ε to 1.0, the window length w to 5, the graph size $|V_{\mathbf{G}}|$ to $|V|$, and the number of timestamps $T_{\mathbf{G}}$ to T. For O-ODLC, we use a step size of $\theta = 15$ and allocate the privacy budget according to $\varepsilon_1 : \varepsilon_2 : \varepsilon_3 : \varepsilon_4 = 1 : 1 : 1 : 7$.

Choice of $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$. Following [46], we assume that $\deg_{\max}^+$ and $\deg_{\max}^-$ are public, and set the projection thresholds $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$ to $\lceil \deg_{\max}^+ \rceil$ and $\lceil \deg_{\max}^- \rceil$, respectively.

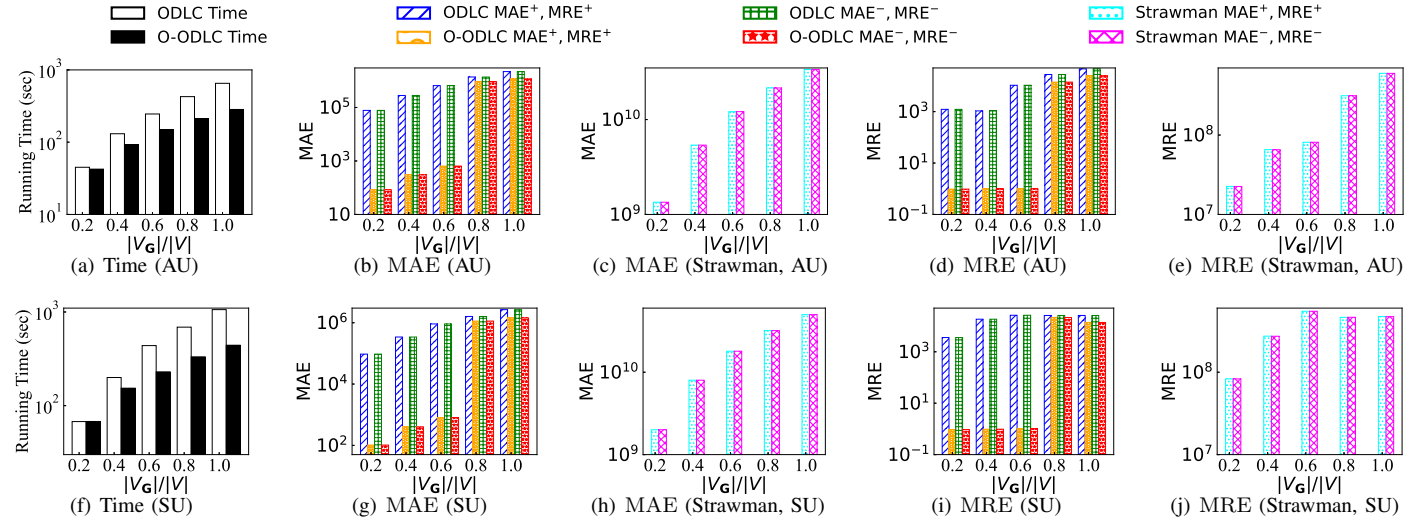
Utility Measures. To assess the utility of the proposed and baseline methods, we report two metrics: mean absolute error (MAE) and mean relative error (MRE). Let $T_{\mathbf{G}}$ be the number of timestamps. For any timestamp $t \in [T_{\mathbf{G}}]$, let (d_t^+, d_t^-) denote the ground-truth degree list of snapshot at timestamp t and $(\overline{d}_t^+, \overline{d}_t^-)$ denote the output degree list. Then, we use these metrics: (1) $\text{MAE}^+ = \frac{1}{T_{\mathbf{G}}} \sum_{t=1}^{T_{\mathbf{G}}} \|d_t^+ - \overline{d}_t^+\|_1$; (2) $\text{MAE}^- = \frac{1}{T_{\mathbf{G}}} \sum_{t=1}^{T_{\mathbf{G}}} \|d_t^- - \overline{d}_t^-\|_1$; (3) $\text{MRE}^+ = \frac{1}{T_{\mathbf{G}}} \sum_{t=1}^{T_{\mathbf{G}}} \frac{\|d_t^+ - \overline{d}_t^+\|_1}{\|d_t^+\|_1}$; (4)



(a) Time (AU) (b) MAE (AU) (c) MAE (Strawman, AU) (d) MRE (AU) (e) MRE (Strawman, AU)

(f) Time (SU) (g) MAE (SU) (h) MAE (Strawman, SU) (i) MRE (SU) (j) MRE (Strawman, SU)

Fig. 6: Impact of window length w



(a) Time (AU) (b) MAE (AU) (c) MAE (Strawman, AU) (d) MRE (AU) (e) MRE (Strawman, AU)

(f) Time (SU) (g) MAE (SU) (h) MAE (Strawman, SU) (i) MRE (SU) (j) MRE (Strawman, SU)

Fig. 7: Impact of graph size $|V_{\mathbf{G}}|$

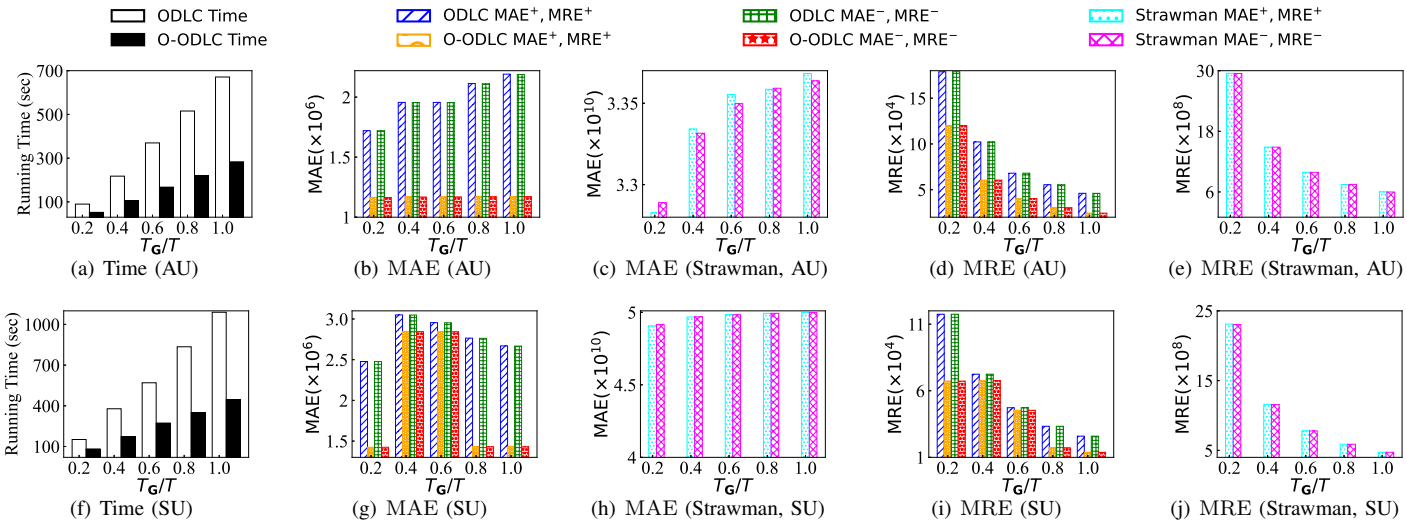
$$\mathrm{MRE}^{-} = \frac{1}{T_{\mathbf{G}}} \sum_{t=1}^{T_{\mathbf{G}}} \frac{\|d_t^{-} - \bar{d}_t^{-}\|_1}{\|d_t^{-}\|_1}.$$

B. Experimental Results

Exp-1: Impact of privacy budget ε. Fig. 5 shows that the running times of ODLC and O-ODLC remain stable as the privacy budget ε increases. This is because $|\mathcal{R}|$, the cardinality of the exponential mechanism's output range, does not depend on ε, and the post-processing phase is also independent of ε. These results also indicate that the proposed efficiency optimization technique in Section V-B scales well as ε varies. O-ODLC runs at least twice as fast as ODLC for two main reasons. First, the step size reduces the cardinality of the exponential mechanism's output range $|\mathcal{R}|$. Second, the efficiency optimization avoids updating some users' estimated in-degrees or out-degrees at certain timestamps. In terms of utility, increasing ε reduces MAE^{+}, MAE^{-}, MRE^{+}, and MRE^{-} for both algorithms. Both proposed algorithms also achieve values four orders of magnitude lower than the strawman solution on all four metrics, demonstrating substantially better utility.

Exp-2: Impact of window length w. As shown in Fig. 6, the running times of both ODLC and O-ODLC remain stable as the window length w increases. As in Exp-1, w does not affect the cardinality of the output range $|\mathcal{R}|$ or the post-processing phase. The proposed efficiency optimization technique also scales well with w. The four utility metrics for both algorithms remain stable as w increases, indicating that their utility scales well with the window length. For the strawman solution, MAE^{+} and MAE^{-} also remain stable, while MRE^{+} and MRE^{-} decrease slightly.

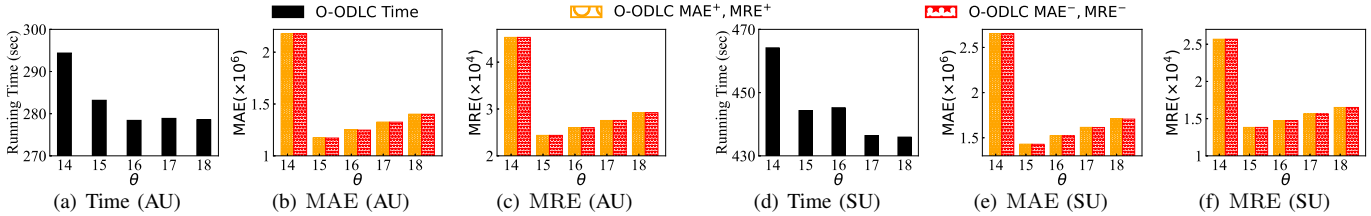
Exp-3: Impact of graph size $|V_{\mathbf{G}}|$. Fig. 7 shows that both ODLC and O-ODLC take longer to run as $|V_{\mathbf{G}}|$ increases, reflecting the larger input sizes. Both algorithms nevertheless scale well in terms of efficiency. The four utility metrics also increase for ODLC and O-ODLC as the graph grows. This increase does not by itself indicate a loss of utility: the ℓ_1-norms in the metric definitions include more terms as the number of vertices increases. In fact, the utility metrics scale well with graph size $|V_{\mathbf{G}}|$. The strawman solution's four metrics also increase with graph size $|V_{\mathbf{G}}|$.



(a) Time (AU) (b) MAE (AU) (c) MAE (Strawman, AU) (d) MRE (AU) (e) MRE (Strawman, AU)

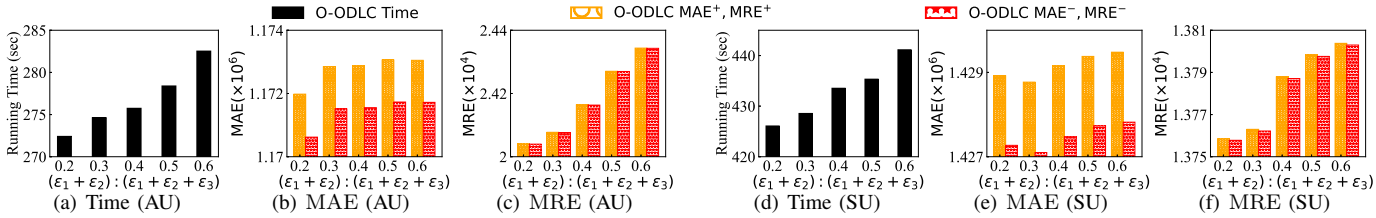
(f) Time (SU) (g) MAE (SU) (h) MAE (Strawman, SU) (i) MRE (SU) (j) MRE (Strawman, SU)

Fig. 8: Impact of the number of timestamps $T_{\mathbf{G}}$



(a) Time (AU) (b) MAE (AU) (c) MRE (AU) (d) Time (SU) (e) MAE (SU) (f) MRE (SU)

Fig. 9: Impact of step size θ



(a) Time (AU) (b) MAE (AU) (c) MRE (AU) (d) Time (SU) (e) MAE (SU) (f) MRE (SU)

Fig. 10: Impact of privacy budget allocation $(\varepsilon_1 + \varepsilon_2) : (\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$

Exp-4: Impact of the number of timestamps $T_{\mathbf{G}}$. Fig. 8 shows that both ODLC and O-ODLC take longer to run as the number of timestamps $T_{\mathbf{G}}$ increases, since processing more timestamps requires more computation. For dataset AU, as $T_{\mathbf{G}}$ increases, MAE^+ and MAE^- for ODLC increase, whereas MRE^+ and MRE^- decrease. This difference reflects the fact that MAE measures absolute error, whereas MRE measures error relative to the degree magnitudes; the two metrics can therefore exhibit different trends as the stream grows.

Exp-5: Impact of step size θ. Fig. 9 shows that the running time of O-ODLC decreases as θ increases. A larger step size reduces the cardinality of the output range $\mathcal{R}$, so the exponential mechanism evaluates fewer candidates. The four utility metrics for O-ODLC first decrease and then increase as θ grows. For small values of θ, reducing $|\mathcal{R}|$ alleviates sparsity in the output distribution and improves utility. Once θ exceeds a threshold, however, the output range becomes too small to support accurate degree estimation.

Exp-6: Impact of privacy budget allocation $(\varepsilon_1 + \varepsilon_2) : (\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$. We examine the privacy budget allocation ratio $(\varepsilon_1 + \varepsilon_2) : (\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$ used by the efficiency optimization technique. Fig. 10 shows that O-ODLC's running time increases as this ratio grows. With the total budget $(\varepsilon_1+\varepsilon_2+\varepsilon_3)$ fixed, assigning a larger budget to the thresholds, that is, increasing $(\varepsilon_1+\varepsilon_2)$, causes more degrees to be updated through the exponential mechanism at each timestamp. The four utility metrics also increase with the ratio, mean that updating more degrees through the exponential mechanism introduces greater error than updating fewer.

VII. Conclusion

In this paper, we study the locally differentially private degree list collection problem in streaming directed graphs. To solve this problem, we propose two algorithms, ODLC algorithm and O-ODLC algorithm. ODLC algorithm is based on a novel composition theorem. O-ODLC algorithm overcomes the sparsity problem of the exponential mechanism and uses a novel efficiency optimization technique to utilize the correlation of different timestamp snapshots. We also provide a novel post processing algorithm based on a discrete mathematics theorem. Experimental results show that the proposed algorithms have high efficiency and utility.

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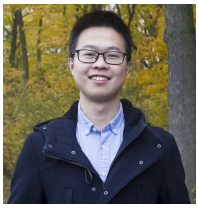
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Ziyao Wei received the B.Eng. degree in computer science from Shandong University, China, in 2022. He is currently working toward the PhD degree with Zhejiang University, China. His research interests include differential privacy and secure graph analysis.



Qing Liu received the PhD degree from the College of Computer Science, Zhejiang University, China. He is currently a professor with the College of Computer Science, Zhejiang University. His research interests include graph management, database usability, and secure graph analysis.



Zhikun Zhang (Member, IEEE) received the PhD degree from Zhejiang University in 2019. From 2017 to 2019, he was a visiting scholar with Purdue University. He also was a visiting assistant professor with Stanford University, research group leader with CISPA Helmholtz Center for Information Security, Germany, and postdoc research with CISPA for two years. He is currently an assistant professor with the College of Computer Science and Technology, Zhejiang University. His research interests include privacy computing, data privacy protection, machine learning privacy, and security. He has been granted with National Science Fund Program for Distinguished Young Scholars (Overseas).



Yunjun Gao (Senior Member, IEEE) received the PhD degree in computer science from Zhejiang University, China, in 2008. He is currently a professor with the College of Computer Science, Zhejiang University. His research interests include database, Big Data management and analytics, AI interaction with DB technology.

APPENDIX A
PROOF OF THEOREM IV.1

For any two w-neighboring neighbor lists $\mathbf{u}$ and $\mathbf{u}'$, for each $i \in [w]$, let $D_i(\mathbf{u})$ and $D_i(\mathbf{u}')$ be the elements of the algorithm $\mathcal{A}_i$ depends on. For any measurable subset $\mathcal{O} \subseteq \mathrm{Range}(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w)$,

$$\begin{aligned}
&\Pr[(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w)(\mathbf{u}) \in \mathcal{O}] \\
=&\prod_{i=1}^{w} \Pr[\mathcal{A}_i(D_1(\mathbf{u}), \cdots, D_i(\mathbf{u}), \cdots, D_w(\mathbf{u}))) \in \mathcal{O}_i] \\
\leq&\prod_{i=1}^{w} e^{\varepsilon_i} \Pr[\mathcal{A}_i(D_1(\mathbf{u}'), \cdots, D_i(\mathbf{u}'), \cdots, D_w(\mathbf{u}'))) \in \mathcal{O}_i] \\
=&e^{\sum_{i=1}^{w} \varepsilon_i} \Pr[(\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_w)(\mathbf{u}') \in \mathcal{O}].
\end{aligned}$$

Here, $\mathcal{O}_i = \pi_i(\mathcal{O})$. π_i denotes the projection mapping.

APPENDIX B
PROOF OF PROPOSITION IV.2

Let n be the number of users. $\forall i \in [n]$,

$$\begin{aligned}
GS_{f_i^+} &= \max_{r \in [0, n-1]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in W} |f_i^+(\boldsymbol{\lambda}, r) - f_i^+(\boldsymbol{\mu}, r)| \\
&= \max_{r \in [0, n-1]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in W} ||\sum_{j=1}^{n} \lambda_j - r| - |\sum_{j=1}^{n} \mu_j - r|| \\
&= \max_{r \in [0, n-1]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in W} \max\{|\sum_{j=1}^{n} \lambda_j - r|, |\sum_{j=1}^{n} \mu_j - r|\} \\
&= \max_{r \in [0, n-1]} \max\{\max_{\boldsymbol{\lambda} \in W} |\sum_{j=1}^{n} \lambda_j - r|, \max_{\boldsymbol{\mu} \in W} |\sum_{j=1}^{n} \mu_j - r|\} \\
&= \max_{r \in [0, n-1]} \max\{r, n-1-r\} \\
&= n-1.
\end{aligned}$$

Here $W = \{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\}$. Similarly, it is easy to calculate that $GS_{f_i^-} = n-1$.

APPENDIX C
PROOF OF PROPOSITION IV.3

Let n be the number of users. $\forall i \in [n]$,

$$\begin{aligned}
GS_{\tilde{f}_i^+} &= \max_{r \in [0, \tilde{d}_{\max}^+]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in W} |\tilde{f}_i^+(\boldsymbol{\lambda}, r) - \tilde{f}_i^+(\boldsymbol{\mu}, r)| \\
&= \max_{r \in [0, \tilde{d}_{\max}^+]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in \tilde{W}} ||\sum_{j=1}^{n} \lambda_j - r| - |\sum_{j=1}^{n} \mu_j - r|| \\
&= \max_{r \in [0, \tilde{d}_{\max}^+]} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu} \in \tilde{W}} \max\{|\sum_{j=1}^{n} \lambda_j - r|, |\sum_{j=1}^{n} \mu_j - r|\} \\
&= \max_{r \in [0, \tilde{d}_{\max}^+]} \max\{\max_{\boldsymbol{\lambda} \in \tilde{W}} |\sum_{j=1}^{n} \lambda_j - r|, \max_{\boldsymbol{\mu} \in \tilde{W}} |\sum_{j=1}^{n} \mu_j - r|\} \\
&= \max_{r \in [0, \tilde{d}_{\max}^+]} \max\{r, \tilde{d}_{\max}^+ - r\} \\
&= \tilde{d}_{\max}^+.
\end{aligned}$$

Here, $W = \{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\}$ and $\tilde{W} = \{\mathbf{a} \in \{0,1\}^n : \sum_{j=1}^{n} a_j \leq \tilde{d}_{\max}^+, a_i \neq 1\}$. Similarly, it is easy to calculate that $GS_{\tilde{f}_i^-} = \tilde{d}_{\max}^-$.

Algorithm 4: Degree Collection

Input : G_t represented as $\mathbf{u}_{1,t}$, $\mathbf{u}_{2,t}$, $\cdots$, $\mathbf{u}_{n,t}$; the thresholds $\tilde{d}_{\max}^+$ and $\tilde{d}_{\max}^-$; the number of users n; privacy budget $\varepsilon \in \mathbb{R}_{\geq 0}$

Output: The estimated out-degrees $\hat{d}_{t,i}^+$ and in-degrees $\hat{d}_{t,i}^-$ of user v_i on v_i's side, $\forall i \in [n]$

```
// On each user side
```

1 **for** $i = 1$ *to* n **do**
2 $\quad d_{t,i}^+ \leftarrow \sum_{j=1}^{n} u_{i,t,j}^+$;
3 $\quad d_{t,i}^- \leftarrow \sum_{j=1}^{n} u_{i,t,j}^-$;
4 $\quad$ **for** $j = 0$ *to* $\tilde{d}_{\max}^+$ **do**
5 $\quad\quad p^+(j) \leftarrow \exp(-\frac{\varepsilon |\min\{d_{t,i}^+, \tilde{d}_{\max}^+\} - j|}{2\tilde{d}_{\max}^+})$;
6 $\quad p_{\mathrm{sum}}^+ \leftarrow \sum_{j=0}^{\tilde{d}_{\max}^+} p^+(j)$;
7 $\quad$ Sample $\hat{d}_{t,i}^+$ from $\mathrm{Categorical}(\frac{1}{p_{\mathrm{sum}}^+}(p^+(0), p^+(1), \cdots, p^+(\tilde{d}_{\max}^+)))$;
8 $\quad$ **for** $j = 0$ *to* $\tilde{d}_{\max}^-$ **do**
9 $\quad\quad p^-(j) \leftarrow \exp(-\frac{\varepsilon |\min\{d_{t,i}^-, \tilde{d}_{\max}^-\} - j|}{2\tilde{d}_{\max}^-})$;
10 $\quad p_{\mathrm{sum}}^- \leftarrow \sum_{j=0}^{\tilde{d}_{\max}^-} p^-(j)$;
11 $\quad$ Sample $\hat{d}_{t,i}^-$ from $\mathrm{Categorical}(\frac{1}{p_{\mathrm{sum}}^-}(p^+(0), p^+(1), \cdots, p^+(\tilde{d}_{\max}^-)))$;
12 $\quad$ Report $\hat{d}_{t,i}^+$ and $\hat{d}_{t,i}^-$ to the central server;

```
// On the central sever side
```

13 Aggregate $\hat{d}_t^+ = (\hat{d}_{t,1}^+, \hat{d}_{t,2}^+, \cdots, \hat{d}_{t,n}^+)$;
14 Aggregate $\hat{d}_t^- = (\hat{d}_{t,1}^-, \hat{d}_{t,2}^-, \cdots, \hat{d}_{t,n}^-)$;
15 **return** $\hat{d}_t = (\hat{d}_t^+, \hat{d}_t^-)$;

APPENDIX D
DEGREE COLLECTION ALGORITHM

As shown in Algorithm 4, the algorithm first runs on each user side. For each user v_i, v_i first obtains the out-degree $d_{t,i}^+$ and in-degree $d_{t,i}^-$, respectively (Lines 2, 3). Then, user v_i implements the exponential mechanism. For all $j \in [0, \tilde{d}_{\max}^+]$, v_i computes the weight $p^+(j)$ and then obtains p_{sum}^+ by summing all $p^+(j)$. (Lines 4-6) Then, the algorithm obtains the estimated out-degree $\hat{d}_{t,i}^+$ sampled from the categorical distribution generated by all the weights (Line 7). Next, user v_i implements the exponential mechanism, once again. For all $j \in [0, \tilde{d}_{\max}^-]$, v_i computes the weight $p^-(j)$ and then obtains p_{sum}^+ by summing all $p^-(j)$. (Lines 8-10) Then, the algorithm obtains the estimated in-degree $\hat{d}_{t,i}^-$ sampled from the categorical distribution generated by all the weights (Line 11). After obtaining the estimated out-degree $\hat{d}_{t,i}^+$ and in-degree $\hat{d}_{t,i}^-$, user v_i reports them to the central server (Line 12). Next, the algorithm runs on the central server side, and the central server aggregates all the estimated out-degrees and in-degrees reported from each user and obtains the estimated degree list $\hat{d}_t = (\hat{d}_t^+, \hat{d}_t^-)$ (Lines 13-15).

APPENDIX E
PROOF OF THEOREM IV.7

Let $\mathcal{A}_t$ be the sub-algorithm of Algorithm IV.7 running at timestamp t, $\forall t \in [T]$. By Theorem III.4, Theorem III.8, and Theorem III.9, $\mathcal{A}_t$ satisfies 1-event local differential privacy for each user with privacy budget ε/w. Then, for any window $\mathcal{I} \in \{[s, s+w-1] : s \in [T-w+1]\}$, by Theorem IV.1,

the composition $(\mathcal{A}_t)_{t\in\mathcal{I}}$ satisfies w-event local differential privacy for each user with privacy budget ε. Next, let n be the number of users. $\forall i \in [n]$, for any pair of w-neighboring neighbor lists $\mathbf{u}_i$ and $\mathbf{u}'_i$ of user v_i, take any interval $\mathcal{J}_i \subseteq [T]$, such that $|\mathcal{J}_i| = w$ and $\mathcal{J}_i \supseteq \{t \in [T] : \mathbf{u}_{i,t} \neq \mathbf{u}'_{i,t}\}$. Then, for any subset $\mathcal{O} \subseteq \mathrm{Range}((\mathcal{A}_t)_{t\in[T]})$,

$$\begin{aligned}
&\Pr[(\mathcal{A}_t)_{t\in[T]}(\mathbf{u}_i) \in \mathcal{O}]\\
=&\Pr[(\mathcal{A}_t)_{t\in\mathcal{J}_i}(\mathbf{u}_i) \in \mathcal{O}_{\mathcal{J}_i}, (\mathcal{A}_t)_{t\in[T]\setminus\mathcal{J}_i}(\mathbf{u}_i) \in \mathcal{O}_{\overline{\mathcal{J}_i}}]\\
=&\Pr[(\mathcal{A}_t)_{t\in\mathcal{J}_i}(\mathbf{u}_i) \in \mathcal{O}_{\mathcal{J}_i}] \cdot \Pr[(\mathcal{A}_t)_{t\in[T]\setminus\mathcal{J}_i}(\mathbf{u}_i) \in \mathcal{O}_{\overline{\mathcal{J}_i}}]\\
\leq&e^{\varepsilon}\Pr[(\mathcal{A}_t)_{t\in\mathcal{J}_i}(\mathbf{u}'_i) \in \mathcal{O}_{\mathcal{J}_i}] \cdot \Pr[(\mathcal{A}_t)_{t\in[T]\setminus\mathcal{J}_i}(\mathbf{u}'_i) \in \mathcal{O}_{\overline{\mathcal{J}_i}}]\\
=&e^{\varepsilon}\Pr[(\mathcal{A}_t)_{t\in[T]}(\mathbf{u}'_i) \in \mathcal{O}].
\end{aligned}$$

Here, $O_{\mathcal{J}_i} = \pi_{\mathcal{J}_i} O$ and $O_{\overline{\mathcal{J}_i}} = \pi_{[T]\setminus\mathcal{J}_i} O$. π denotes the projection operator. Therefore, according to Definition III.2, Algorithm 2 satisfies w-event local differential privacy for each user with privacy budget ε.

APPENDIX F
PROOF OF PROPOSITION V.1

This proof is by contradiction. Suppose that $|A_\delta| \geq \frac{1}{\delta}$. Then,

$$\mathbb{P}(\Omega) = \sum_{x\in\Omega} \mathbb{P}(\{x\}) = \sum_{x\in A_\delta} \mathbb{P}(\{x\}) > \delta \cdot \frac{1}{\delta} = 1.$$

This contradicts $\mathbb{P}(\Omega) = 1$.

APPENDIX G
PROOF OF PROPOSITION V.2

Let n be the number of users. $\forall i \in [n]$,

$$\begin{aligned}
GS_{\tilde{f}^+_{i,\theta}} &= \max_{r\in U} \max_{\boldsymbol{\lambda},\boldsymbol{\mu}\in W} |\tilde{f}^+_{i,\theta}(\boldsymbol{\lambda}, r) - \tilde{f}^+_{i,\theta}(\boldsymbol{\mu}, r)|\\
&= \max_{r\in U} \max_{\boldsymbol{\lambda},\boldsymbol{\mu}\in \tilde{W}} ||\sum_{j=1}^{n} \lambda_j - r| - |\sum_{j=1}^{n} \mu_j - r||\\
&= \max_{r\in U} \max_{\boldsymbol{\lambda},\boldsymbol{\mu}\in \tilde{W}} \max\{|\sum_{j=1}^{n} \lambda_j - r|, |\sum_{j=1}^{n} \mu_j - r|\}\\
&= \max_{r\in U} \max\{\max_{\boldsymbol{\lambda}\in \tilde{W}} |\sum_{j=1}^{n} \lambda_j - r|, \max_{\boldsymbol{\mu}\in \tilde{W}} |\sum_{j=1}^{n} \mu_j - r|\}\\
&= \max_{r\in U} \max\{r, \tilde{d}^+_{\max} - r\}\\
&= \tilde{d}^+_{\max}.
\end{aligned}$$

Here, $U = \{j\theta : 0 \leq j \leq \tilde{d}^+_{\max}/\theta, j \in \mathbb{Z}\}$, $W = \{\mathbf{a} \in \{0,1\}^n : a_i \neq 1\}$, and $\tilde{W} = \{\mathbf{a} \in \{0,1\}^n : \sum_{j=1}^{n} a_j \leq \tilde{d}^+_{\max}, a_i \neq 1\}$. Similarly, it is easy to calculate that $GS_{\tilde{f}^-_{i,\theta}} = \tilde{d}^-_{\max}$.

APPENDIX H
PROOF OF THEOREM V.3

Let $\mathcal{A}^+_t$ and $\mathcal{A}^-_t$ denote the determination processes of Algorithm 3 at each timestamp t. Specially, $\mathcal{A}^+_t$ (resp. $\mathcal{A}^-_t$) denotes the determination process of whether the estimated out-degree (resp. estimated in-degree) should be updated. We first prove that both $\mathcal{A}^+_t$ and $\mathcal{A}^-_t$ satisfy w-event local differential privacy for each user with privacy budget $(\varepsilon_1+\varepsilon_2+\varepsilon_3)/w$.

For each user v_i, consider two w-neighboring neighbor lists $\mathbf{u}_i$ and $\mathbf{u}'_i$, with the corresponding projected out-degrees $\tilde{d}$ and $\tilde{d}'$, respectively. Notice that $|\tilde{d}-\tilde{d}'| \leq \tilde{d}^+_{\max}$ since the neighbor list projection. Let

$$\begin{aligned}
g(\mathbf{a}, x, y) &= \Pr[(\tilde{d} + \xi \leq y) \vee (\tilde{d} + \xi \geq x)],\\
h(\mathbf{a}, x, y) &= \Pr[y < \tilde{d} + \xi < x].
\end{aligned}$$

Here, d corresponds to the number of 1 elements in $\mathbf{a}$ after the neighbor list projection. Notice that η^+_u and η^+_l respectively have probability density functions,

$$\begin{aligned}
f_{\eta^+_u}(x) &= \frac{\varepsilon_1}{2w\tilde{d}^+_{\max}} \exp\left(-\frac{\varepsilon_1|x|}{w\tilde{d}^+_{\max}}\right),\\
f_{\eta^+_l}(y) &= \frac{\varepsilon_2}{2w\tilde{d}^+_{\max}} \exp\left(-\frac{\varepsilon_2|y|}{w\tilde{d}^+_{\max}}\right).
\end{aligned}$$

The determination has two possible results, (1) update (denoted as $\top$), (2) maintain (denoted as $\perp$). For any $o \in \{\top, \perp\}$, let

$$\phi(\mathbf{a}, x, y) = \begin{cases} g(\mathbf{a}, x, y), & \text{if } o = \top,\\ h(\mathbf{a}, x, y), & \text{if } o = \perp. \end{cases}$$

Then,

$$\begin{aligned}
&\Pr[\mathcal{A}^+_t(\mathbf{u}_i) = o] = \int_{\mathbb{R}^2} f_{\eta^+_u}(x) f_{\eta^+_l}(y) \phi(\mathbf{u}_i, x, y) \mathrm{d}x\mathrm{d}y\\
=&\int_{\mathbb{R}^2} f_{\eta^+_u}(x - \tilde{d}^+_{\max}) f_{\eta^+_l}(y + \tilde{d}^+_{\max}) \phi(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max}) \mathrm{d}x\mathrm{d}y,\\
&\Pr[\mathcal{A}^+_t(\mathbf{u}'_i) = o] = \int_{\mathbb{R}^2} f_{\eta^+_u}(x) f_{\eta^+_l}(y) \phi(\mathbf{u}'_i, x, y) \mathrm{d}x\mathrm{d}y.
\end{aligned}$$

Notice that,

$$\begin{aligned}
f_{\eta^+_u}(x - \tilde{d}^+_{\max}) &\leq e^{\varepsilon_1/w} f_{\eta^+_u}(x),\\
f_{\eta^+_l}(y + \tilde{d}^+_{\max}) &\leq e^{\varepsilon_2/w} f_{\eta^+_l}(y).
\end{aligned}$$

Then, (1) if $o = \top$. Let $|\tilde{d} - \tilde{d}'| = \Delta$. (i) If $\tilde{d} \leq \tilde{d}'$,

$$\begin{aligned}
&\phi(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max})\\
=&g(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max})\\
=&\Pr[(\tilde{d} + \xi \leq y + \tilde{d}^+_{\max}) \vee (\tilde{d} + \xi \geq x - \tilde{d}^+_{\max})]\\
=&\Pr[\tilde{d} + \xi \leq y + \tilde{d}^+_{\max}] + \Pr[\tilde{d} + \xi \geq x - \tilde{d}^+_{\max}]\\
&- \Pr[x - \tilde{d}^+_{\max} \leq \tilde{d} + \xi \leq y + \tilde{d}^+_{\max}]\\
=&\Pr[\tilde{d} + \xi \leq y + \tilde{d}^+_{\max}] + \Pr[\tilde{d} + \xi \geq \max\{x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max}\}],
\end{aligned}$$

and

$$\begin{aligned}
&\phi(\mathbf{u}'_i, x, y)\\
=&g(\mathbf{u}'_i, x, y)\\
=&\Pr[(\tilde{d}' + \xi \leq y) \vee (\tilde{d}' + \xi \geq x)]\\
=&\Pr[(\tilde{d} + \xi \leq y - \Delta) \vee (\tilde{d} + \xi \geq x - \Delta)]\\
=&\Pr[\tilde{d} + \xi \leq y - \Delta] + \Pr[\tilde{d} + \xi \geq x - \Delta]\\
&- \Pr[x - \Delta \leq \tilde{d} + \xi \leq y - \Delta]\\
=&\Pr[\tilde{d} + \xi \leq y - \Delta] + \Pr[\tilde{d} + \xi \geq \max\{x - \Delta, y - \Delta\}].
\end{aligned}$$

Since ξ follows the Laplace distribution with location param-

eter 0 and scale parameter $\frac{2w\tilde{d}^+_{max}}{\varepsilon_3}$,

$$\begin{aligned}\Pr[\tilde{d}+\xi \le y+\tilde{d}^+_{\max}] &\le e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \le y-\tilde{d}^+_{\max}]\\ &\le e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \le y-\Delta].\end{aligned}$$

Comparing $\max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}$ with $\max\{x-\Delta, y-\Delta\}$, if $\max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} \ge \max\{x-\Delta, y-\Delta\}$, we have,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \ge \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&\Pr[\tilde{d}+\xi \ge \max\{x-\Delta, y-\Delta\}].\end{aligned}$$

Otherwise, $\max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} < \max\{x-\Delta, y-\Delta\}$, then

$$\max\{x-\Delta, y-\Delta\} = x-\Delta.$$

If $\max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} = x-\tilde{d}^+_{\max}$, then

$$\begin{aligned}&\max\{x-\Delta, y-\Delta\} - \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}\\ =&\tilde{d}^+_{\max}-\Delta.\end{aligned}$$

Otherwise, $\max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} = y+\tilde{d}^+_{\max}$, then

$$\begin{aligned}&\max\{x-\Delta, y-\Delta\} - \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}\\ =&x-y-\Delta-\tilde{d}^+_{\max} \le \tilde{d}^+_{\max}-\Delta.\end{aligned}$$

Therefore,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \ge \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \ge \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}+2\tilde{d}^+_{\max}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \ge \max\{x-\Delta, y-\Delta\}].\end{aligned}$$

Thus, we have,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \ge \max\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \ge \max\{x-\Delta, y-\Delta\}].\end{aligned}$$

Furthermore,

$$\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) \le e^{\varepsilon_3/w}\phi(\mathbf{u}'_i, x, y).$$

(ii) Now we turn to the case that $\tilde{d} \ge \tilde{d}'$.

$$\begin{aligned}&\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max})\\ =&g(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max})\\ =&\Pr[(\tilde{d}+\xi \le y+\tilde{d}^+_{\max}) \vee (\tilde{d}+\xi \ge x-\tilde{d}^+_{\max})]\\ =&\Pr[\tilde{d}+\xi \le y+\tilde{d}^+_{\max}] + \Pr[\tilde{d}+\xi \ge x-\tilde{d}^+_{\max}]\\ &-\Pr[x-\tilde{d}^+_{\max} \le \tilde{d}+\xi \le y+\tilde{d}^+_{\max}]\\ =&\Pr[\tilde{d}+\xi \ge x-\tilde{d}^+_{\max}]\\ &+\Pr[\tilde{d}+\xi \le \min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}],\end{aligned}$$

and

$$\begin{aligned}&\phi(\mathbf{u}'_i, x, y) = g(\mathbf{u}'_i, x, y)\\ =&\Pr[(\tilde{d}'+\xi \le y) \vee (\tilde{d}'+\xi \ge x)]\\ =&\Pr[(\tilde{d}+\xi \le y+\Delta) \vee (\tilde{d}+\xi \ge x+\Delta)]\\ =&\Pr[\tilde{d}+\xi \le y+\Delta] + \Pr[\tilde{d}+\xi \ge x+\Delta]\\ &-\Pr[x+\Delta \le \tilde{d}+\xi \le y+\Delta]\\ =&\Pr[\tilde{d}+\xi \ge x+\Delta] + \Pr[\tilde{d}+\xi \le \min\{x+\Delta, y+\Delta\}].\end{aligned}$$

We have,

$$\begin{aligned}\Pr[\tilde{d}+\xi \ge x-\tilde{d}^+_{\max}] &\le e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \ge x+\tilde{d}^+_{\max}]\\ &\le e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \ge x+\Delta].\end{aligned}$$

Comparing $\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}$ with $\min\{x+\Delta, y+\Delta\}$, if $\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} \le \min\{x+\Delta, y+\Delta\}$,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \le \min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&\Pr[\tilde{d}+\xi \le \min\{x+\Delta, y+\Delta\}].\end{aligned}$$

Otherwise, $\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} > \min\{x+\Delta, y+\Delta\}$, then

$$\min\{x+\Delta, y+\Delta\} = y+\Delta.$$

If $\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} = x-\tilde{d}^+_{\max}$, then

$$\begin{aligned}&\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} - \min\{x+\Delta, y+\Delta\}\\ =&x-y-\Delta-\tilde{d}^+_{\max} \le \tilde{d}^+_{\max}-\Delta.\end{aligned}$$

Otherwise, $\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} = y+\tilde{d}^+_{\max}$, then

$$\min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\} - \min\{x+\Delta, y+\Delta\} = \tilde{d}^+_{\max}-\Delta.$$

Therefore,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \le \min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \le \min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}-2\tilde{d}^+_{\max}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \le \min\{x+\Delta, y+\Delta\}].\end{aligned}$$

Thus, we have,

$$\begin{aligned}&\Pr[\tilde{d}+\xi \le \min\{x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}\}]\\ \le&e^{\varepsilon_3/w}\Pr[\tilde{d}+\xi \le \min\{x+\Delta, y+\Delta\}].\end{aligned}$$

Furthermore,

$$\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) \le e^{\varepsilon_3/w}\phi(\mathbf{u}'_i, x, y).$$

Therefore, for $o = \top$, no matter $\tilde{d} \le \tilde{d}'$ or $\tilde{d}' \le \tilde{d}$,

$$\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) \le e^{\varepsilon_3/w}\phi(\mathbf{u}'_i, x, y).$$

Then, we consider (2) $o = \perp$. Let $|\tilde{d}-\tilde{d}'| = \Delta$, (i) if $\tilde{d} \le \tilde{d}'$,

$$\begin{aligned}\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) =&h(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max})\\ =&\Pr[y+\tilde{d}^+_{\max} < \tilde{d}+\xi < x-\tilde{d}^+_{\max}],\end{aligned}$$

and

$$\begin{aligned}\phi(\mathbf{u}'_i, x, y) =&h(\mathbf{u}'_i, x, y)\\ =&\Pr[y < \tilde{d}'+\xi < x]\\ =&\Pr[y-\Delta < \tilde{d}+\xi < x-\Delta].\end{aligned}$$

It is easy to find that,

$$\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) \le \phi(\mathbf{u}'_i, x, y).$$

(ii) Similarly, for the case that $\tilde{d} > \tilde{d}'$,

$$\begin{aligned}\phi(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max}) =&h(\mathbf{u}_i, x-\tilde{d}^+_{\max}, y+\tilde{d}^+_{\max})\\ =&\Pr[y+\tilde{d}^+_{\max} < \tilde{d}+\xi < x-\tilde{d}^+_{\max}],\end{aligned}$$

and

$$\begin{aligned}\phi(\mathbf{u}_i', x, y) =& h(\mathbf{u}_i', x, y)\\ =&\Pr[y < \tilde{d}' + \xi < x]\\ =&\Pr[y + \Delta < \tilde{d} + \xi < x + \Delta].\end{aligned}$$

It is easy to find that,

$$\phi(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max}) \le \phi(\mathbf{u}_i', x, y).$$

Therefore, for $o = \perp$, no matter $\tilde{d} \le \tilde{d}'$ or $\tilde{d}' \le \tilde{d}$,

$$\phi(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max}) \le \phi(\mathbf{u}_i', x, y).$$

Thus,

$$\begin{aligned}&\Pr[\mathcal{A}^+_t(\mathbf{u}_i) = o]\\ =&\int_{\mathbb{R}^2} f_{\eta^+_u}(x - \tilde{d}^+_{\max}) f_{\eta^+_l}(y + \tilde{d}^+_{\max})\\ &\phi(\mathbf{u}_i, x - \tilde{d}^+_{\max}, y + \tilde{d}^+_{\max})\mathrm{d}x\mathrm{d}y\\ \le& e^{(\varepsilon_1+\varepsilon_2+\varepsilon_3)/w}\int_{\mathbb{R}^2} f_{\eta^+_u}(x) f_{\eta^+_l}(y)\phi(\mathbf{u}_i', x, y)\mathrm{d}x\mathrm{d}y\\ =& e^{(\varepsilon_1+\varepsilon_2+\varepsilon_3)/w}\Pr[\mathcal{A}^+_t(\mathbf{u}_i') = o].\end{aligned}$$

Therefore, $\mathcal{A}^+_t$ satisfies 1-event local differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)/w$. Similarly, we can prove that $\mathcal{A}^-_t$ satisfies 1-event local differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)/w$.

Based on this result, at timestamp t, according to Theorem III.4 and Theorem III.7, the sub-algorithm that obtains the estimated out-degrees satisfies 1-event local differential privacy with privacy budget $(\varepsilon_1+\varepsilon_2+\varepsilon_3+\varepsilon_4)/w$ for each user, and the same to the sub-algorithm that obtains the estimated in-degree. Then, by Theorem III.8 and Theorem III.9, the sub-algorithm running at timestamp t satisfies 1-event differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)/w$ for each user. By Theorem IV.1, as the same way of the proof of Theorem IV.7, Algorithm 3 satisfies w-event local differential privacy with privacy budget $(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4)$ for each user.

APPENDIX I
FULL COMPLEXITY ANALYSIS OF ALGORITHM 3

For each timestamp t, on each user v_i's side, Algorithm 3 first determines whether the estimated out-degree should be updated, which takes $O(1)$ time. Then, if the determination is $\perp$, i.e., the estimated out-degree $\hat{d}^+_{t,i}$ retains the prior estimated out-degree $\hat{d}^+_{t-1,i}$, it only takes $O(1)$ time. Otherwise, if the determination is $\top$, i.e., the estimated out-degree $\hat{d}^+_{t,i}$ should be updated via the exponential mechanism. Then, the algorithm uses the exponential mechanism. It should be noted that the step size θ can influence the output range of the exponential mechanism. The algorithm takes $O(\tilde{d}^+_{\max}/\theta)$ time to obtain all the weights, and then takes $O(\tilde{d}^+_{\max}/\theta)$ time to sample the estimated out-degree $\hat{d}^+_{t,i}$ from the categorical distribution. Therefore, at each timestamp t, on each user v_i's side, the algorithm takes at most $O(\tilde{d}^+_{\max}/\theta)$ time to obtain the estimated out-degree. Similarly, the algorithm takes at most $O(\tilde{d}^-_{\max}/\theta)$ time to obtain the estimated in-degree. Thus, at each timestamp t, the algorithm takes $O((\tilde{d}^+_{\max} + \tilde{d}^-_{\max})/\theta)$ time on each user v_i's side. On the central server side, the central server takes $O(n)$ time to aggregate all the reported out-degrees and in-degrees. Recall that the complexity of the post processing algorithm (i.e., Algorithm 1) is $O(n \log n + n\tilde{d}^-_{\max})$. Therefore, the algorithm takes $O(n \log n + n\tilde{d}^-_{\max})$ time on the central server side at each timestamp. Each user reports his/her estimated out-degree and in-degree to the central server. Therefore, the algorithm takes $O(n)$ communication complexity at each timestamp.

APPENDIX J
MISSING EXPERIMENTAL RESULTS

*1) **Exp-1**:* The missing results of **Exp-1** are shown in Fig. 11.

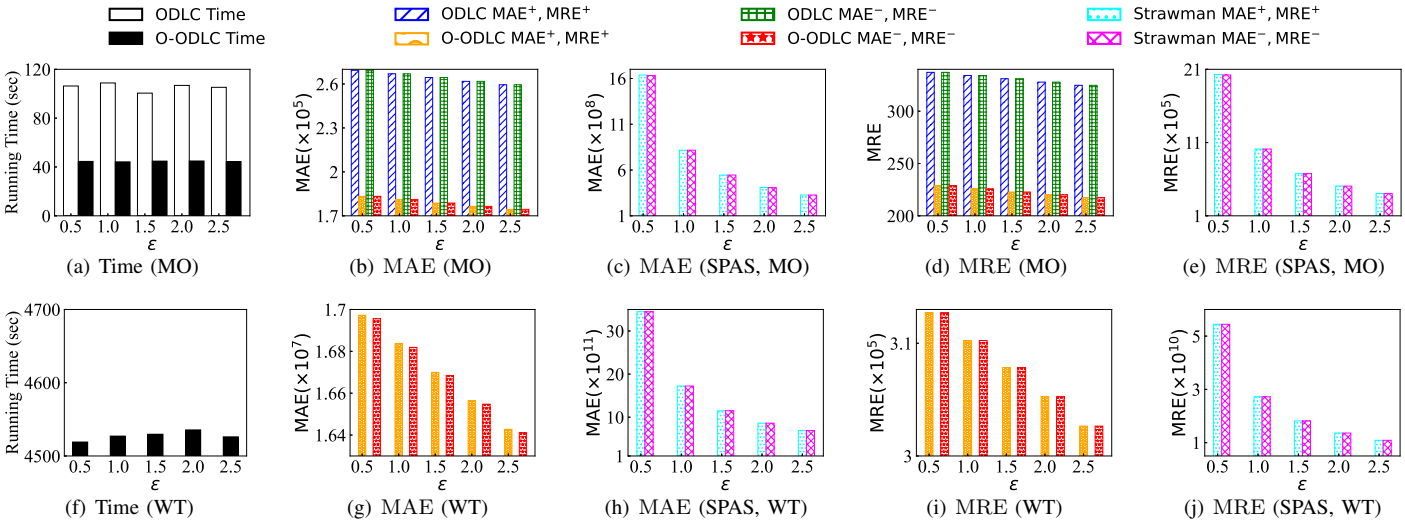
*2) **Exp-2**:* The missing results of **Exp-2** are shown in Fig. 12.

*3) **Exp-3**:* The missing results of **Exp-3** are shown in Fig. 13.

*4) **Exp-4**:* The missing results of **Exp-4** are shown in Fig. 14.

*5) **Exp-5**:* The missing results of **Exp-5** are shown in Fig. 15.

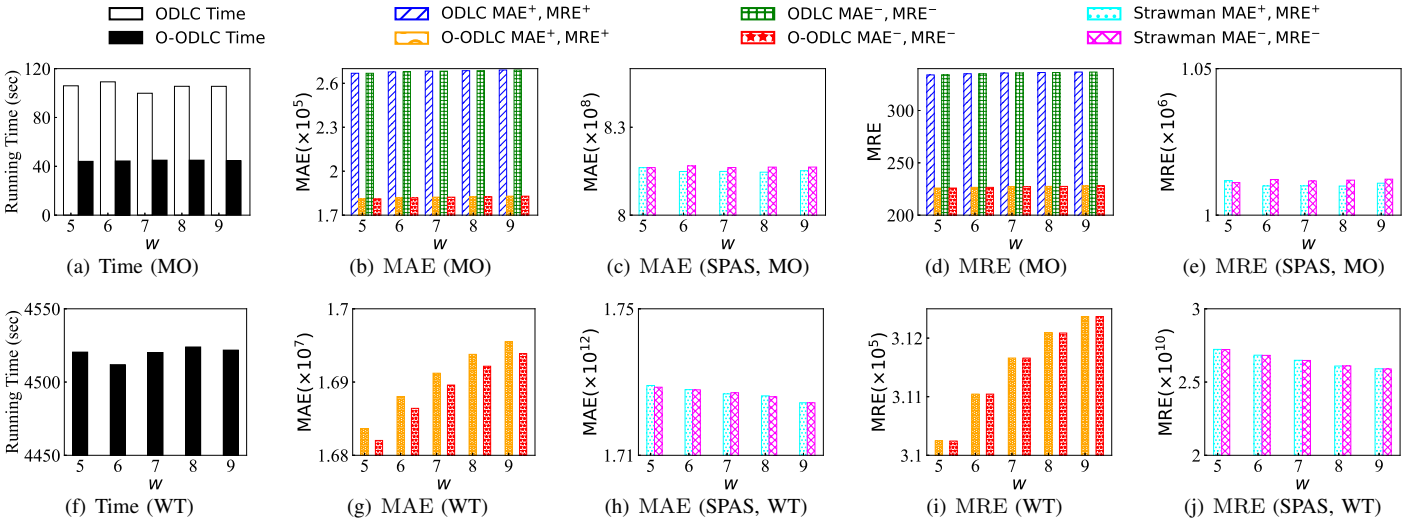
*6) **Exp-6**:* The missing results of **Exp-6** are shown in Fig. 16.



(a) Time (MO) (b) MAE (MO) (c) MAE (SPAS, MO) (d) MRE (MO) (e) MRE (SPAS, MO)

(f) Time (WT) (g) MAE (WT) (h) MAE (SPAS, WT) (i) MRE (WT) (j) MRE (SPAS, WT)

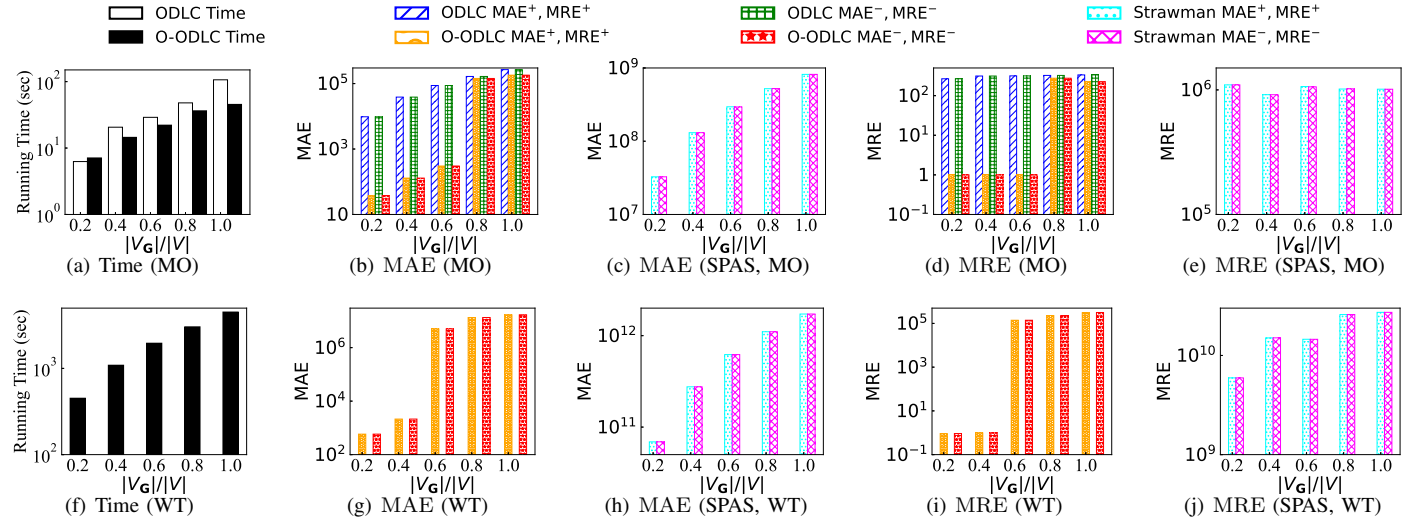
Fig. 11: Impact of privacy budget ε



(a) Time (MO) (b) MAE (MO) (c) MAE (SPAS, MO) (d) MRE (MO) (e) MRE (SPAS, MO)

(f) Time (WT) (g) MAE (WT) (h) MAE (SPAS, WT) (i) MRE (WT) (j) MRE (SPAS, WT)

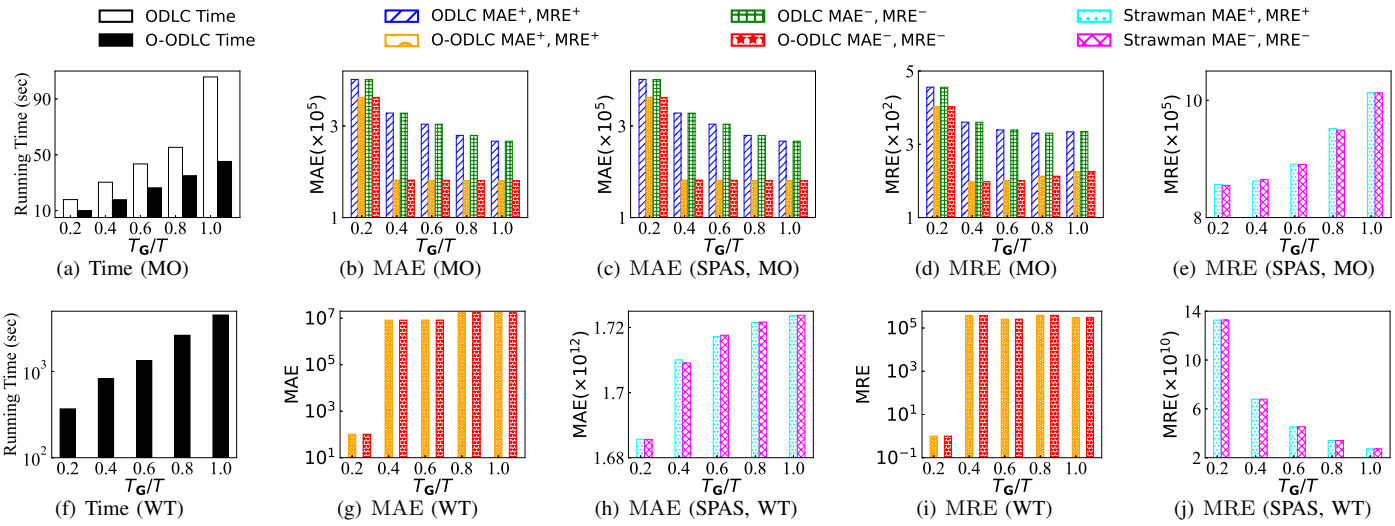
Fig. 12: Impact of window length w



(a) Time (MO) (b) MAE (MO) (c) MAE (SPAS, MO) (d) MRE (MO) (e) MRE (SPAS, MO)

(f) Time (WT) (g) MAE (WT) (h) MAE (SPAS, WT) (i) MRE (WT) (j) MRE (SPAS, WT)

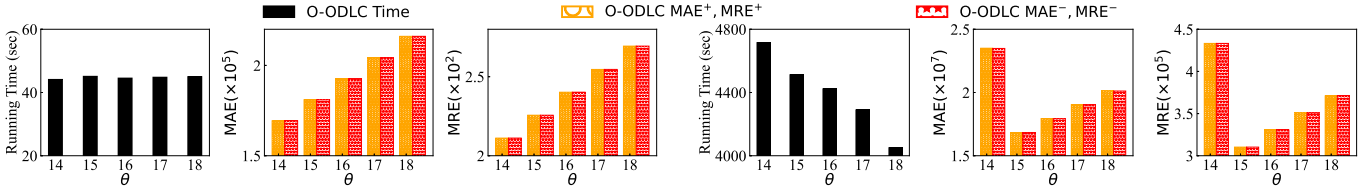
Fig. 13: Impact of graph size $|V_{\mathbf{G}}|$



(a) Time (MO) (b) MAE (MO) (c) MAE (SPAS, MO) (d) MRE (MO) (e) MRE (SPAS, MO)

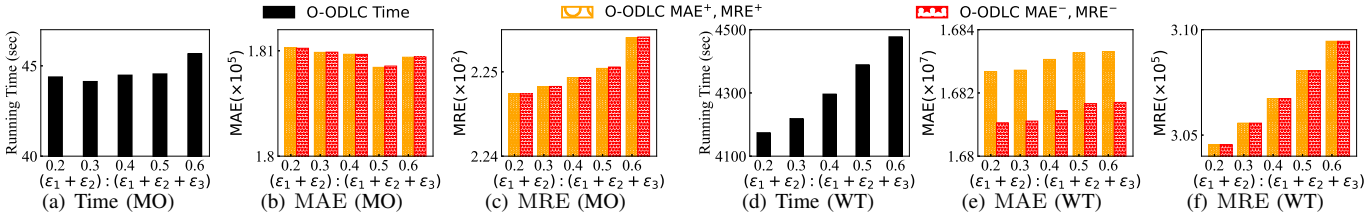
(f) Time (WT) (g) MAE (WT) (h) MAE (SPAS, WT) (i) MRE (WT) (j) MRE (SPAS, WT)

Fig. 14: Impact of the number of timestamps $T_{\mathbf{G}}$



(a) Time (MO) (b) MAE (MO) (c) MRE (MO) (d) Time (WT) (e) MAE (WT) (f) MRE (WT)

Fig. 15: Impact of step size θ



(a) Time (MO) (b) MAE (MO) (c) MRE (MO) (d) Time (WT) (e) MAE (WT) (f) MRE (WT)

Fig. 16: Impact of privacy budget allocation $(\varepsilon_1 + \varepsilon_2) : (\varepsilon_1 + \varepsilon_2 + \varepsilon_3)$